

Mandatory ESG Disclosure and Corporate Hedging: Evidence from Korea

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Abstract

Does mandatory ESG disclosure change how firms hedge? This paper models a firm that jointly chooses hedge ratios and disclosure intensity, with disclosure operating only through the price the firm attaches to residual risk. The model yields closed-form hedge ratios and signed comparative statics, and is tested on Korea's two realized ESG-disclosure mandates (the governance report, phased in by asset tier over 2019–2024, and the environmental-information requirement binding large issuers from 2022) on a 380-firm, 3,420 firm-year panel of KOSPI issuers parsed from OpenDART filings. Group-time ATT estimates are nulls on both hedging margins for both mandates: point estimates within half a percentage point of zero on the primary margin, with confidence intervals that cap a positive response, the sign the competing disclosure channels predict, at one to six percentage points. The channel is risk-specific: mandates that leave market-risk exposure unquantified leave the hedging incentive unchanged. Korea's climate mandate, which compels that quantification from fiscal 2027 with enforcement phasing in through 2030, is the model's dated forward test.

Keywords: mandatory ESG disclosure; corporate hedging; risk management; difference-in-differences; staggered adoption; KSSB; ISSB

JEL classification: G32; M48; Q54; C33

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1 Introduction

Consider a firm holding one economic position: a Korean crude-oil importer’s joint exposure to the WTI price of its physical barrels and the USD/KRW rate at which the dollar liability converts to won. The standard risk-management toolkit interrogates that position from three angles: the static allocation of a fixed premium budget across the two legs, the path-by-path performance of a dynamic delta hedge of a structured position written on the same exposure, and the IFRS 9 hedge-accounting treatment of the resulting relationship. All three take the *decision to hedge*, and its intensity, as a primitive. This paper asks what sets that primitive. Specifically: does a change in the firm’s *disclosure* environment (the arrival of a mandatory ESG and climate-risk reporting regime) change how much of its commodity and currency risk a firm hedges, whether it elects hedge accounting, and how efficiently it converts hedging spend into volatility reduction? The answer the paper builds and then tests, in one line: disclosure moves hedging only through the price the firm puts on un-hedged risk, so a mandate changes hedging only when what it forces into the open is the hedgeable exposure itself. Korea’s two realized mandates fail that condition and produce the zeros of Section 7; the climate mandate that meets it receives a dated, falsifiable forecast.

The question here is a risk-management one with a specific mechanism, distinct from the familiar “does ESG raise firm value” correlation and from work on reporting quality. A disclosure mandate does two things at once to a firm carrying real financial and climate exposure. First, it raises the marginal cost of *leaving risk unhedged*: once a firm must quantify and publish its climate-related financial exposures (KSSB/ISSB S2), residual exposure becomes visible to investors, lenders, and supervisors, and the penalty for carrying it, through cost of capital, covenant scrutiny, and reputational channels, rises. Second, disclosure is itself costly, and the firm chooses how much of it to produce. The firm therefore solves a joint problem in three instruments: how much financial risk to hedge, how much climate risk to hedge, and how much to disclose. The central economic content is that these instruments are coupled (disclosure intensity shifts the shadow price of residual risk, which shifts the optimal hedge ratios) and that the coupling has a sign the data can be asked to confirm.

Korea is an unusually clean laboratory for this question, because it has legislated a sequence of ESG-disclosure mandates that differ in the dimension the model prices: how directly the required disclosure quantifies the firm’s risk exposure. The mandatory corporate-governance report (phased in by asset tier: 2019, 2022, 2024) obliges disclosure about board structure and control (the G pillar) without touching exposure. The environmental-information disclosure requirement (from 2022, for listed firms with assets above KRW 2 trillion) obliges quantitative reporting of emissions and resource use (exposure-adjacent, but not market-risk exposure). The KSSB’s ISSB-aligned sustainability standards (Standards No. 1 and No. 2, mirroring IFRS S1/S2 and adopted in February 2026) oblige quantification of climate-related *financial* exposure itself; the implementation plan finalized in July 2026 (FSC, 2026) schedules them as statutory business-report disclosure under the Capital Markets Act for KOSPI issuers with consolidated assets of KRW 10 trillion or more from fiscal 2027, extending to the 5-trillion tier the following year, with Scope 3 emissions deferred roughly three years and third-party assurance from 2030.

Each mandate phases in by size tier, generating the staggered treatment structure that modern difference-in-differences estimation exploits, with not-yet-covered firms as controls. Korea also has a centralized, machine-readable filing system (the Financial Supervisory Service’s OpenDART) that makes the required firm-level panel constructible from primary filings. One consequence for design should be recorded at the outset: the governance-report requirement extends to all KOSPI issuers from 2026, so the never-treated comparison group for that mandate ceases to exist at the end of the sample used here. Every estimate reported for it is identified off not-yet-treated firms within the phase-in window, and no extension of that panel past 2025 can recover a clean control group.

This paper makes three moves. First (Section 3), it writes down the firm’s joint hedging-and-disclosure problem and solves it: the interior optimum in the two hedge ratios is a closed-form linear-system solution driven by the risk covariance matrix and the premium vector, scaled by a disclosure-dependent effective risk price $\Lambda(d) = \phi + \lambda e^{-kd}$; the disclosure choice is the solution of a scalar first-order condition balancing quadratic disclosure cost against exponential penalty relief. From this the paper derives the comparative statics that the empirics then test: most importantly $\partial h_f / \partial \lambda > 0$ (a tougher penalty regime unambiguously raises hedging), $\partial h_f / \partial a > 0$ (when disclosure is expensive, firms substitute toward hedging), and a closed-form sign for the correlation sensitivity $\partial h_f / \partial \rho$. The disclosure response is subtler: because hedging and disclosure are substitutes in the penalty channel, $\partial d^* / \partial \lambda$ carries the sign of $\phi - \lambda e^{-kd^*}$ (disclosure rises with stringency only when the firm’s own distress-cost motive dominates the marginal regulatory penalty), so “tougher regime, more disclosure” is a conditional prediction.

Second (Sections 4–5), it converts the comparative statics into formal hypotheses and an identification strategy: a heterogeneity-robust staggered DiD with an event-study parallel-trends test, backed by synthetic-control, triple-difference, and instrumental-variable alternatives.

Third (Section 6), it specifies the Korean data pipeline in full (sources, XBRL tags, proxy definitions, winsorization, and missing-data rules) and executes its structured front live against the OpenDART API (Section 6.2), retrieving a real pilot sample and parsing the derivative notes, which shows the hedge-accounting designation reliably extractable but the numeric derivative tables heterogeneous across firms, the binding constraint for full-scale extraction.

This paper pairs a closed-form theory with an executed empirical test. The design was written for the KSSB climate mandate, which has no realized post-period yet; Section 7 therefore runs the design on Korea’s two realized ESG-disclosure mandates, the staggered governance report (2019/2022/2024 by asset tier) and the environmental-information requirement (2022, assets ≥ 2 tn), on a 380-firm, 3,420 firm-year panel built from OpenDART filings. Both mandates produce nulls on every real hedging margin, robust across three comparison-set rules, with placebo estimates indistinguishable from zero, and the nulls discriminate: salience, governance-quality, and compliance-cost channels all predict a hedging response to mandates of this kind, and none appears; a placebo test during construction also identified an encoding artifact in older filings that would otherwise have masqueraded as a positive effect. Read through the model, the nulls identify the penalty channel as risk-specific: neither mandate forces quantification of market-risk exposure, so neither moves the hedging-relevant residual-risk price, and the climate

mandate, which does force it, becomes the standing falsifiable test.

A dedicated numerical engine (Section C) separately checks the solver against an independent optimizer and shows the estimators recover known effects on model-generated panels; Section 8 builds the design around its principal identification threats: endogenous timing, measurement, staggered-adoption bias, and functional-form dependence.

2 Literature Review

This paper sits at the intersection of five literatures: the theory of corporate hedging, the empirics of derivative usage and firm value, hedge accounting under IFRS, the economics of mandatory ESG and climate disclosure, and the econometrics of staggered policy adoption. Table 1 maps the indispensable works in each; the text then states how the present design differs.

Corporate hedging theory. The rationale for costly hedging by value-maximizing firms is the foundation the disclosure question builds on. Smith and Stulz (1985) derive hedging from convexity in taxes, distress costs, and managerial risk aversion; Froot, Scharfstein, and Stein (1993) reframe hedging as the protection of internal funds for investment when external finance is costly, which is the friction that makes a residual-risk penalty economically real and not a modeling device; and Stulz (1996) rationalizes *selective, partial* hedging, the interior $h < 1$ the present model produces. These papers explain *why* $\lambda > 0$; none of them lets the shadow price of residual risk be moved by a disclosure choice, which is this paper’s mechanism. Adler and Dumas (1984) supply the definition of currency exposure the financial leg uses, and Anderson and Danthine (1981) and Ederington (1979) the cross-hedging and minimum-variance-ratio apparatus that the covariance-matrix solution of Section 3 specializes.

Derivative usage and firm value. The empirical hedging literature establishes both the determinants and the measurement of corporate derivative use, and is the direct methodological ancestor of the design here. Nance, Smith, and Smithson (1993) and Géczy, Minton, and Schrand (1997) identify the firm characteristics (size, growth options, financial constraint, foreign exposure) that predict derivative use and that this design carries as controls X_{it} . Tufano (1996) and Jin and Jorion (2006) study commodity producers, the mirror image of the importer position studied here. Allayannis and Weston (2001), Graham and Rogers (2002), Guay and Kothari (2003), and Bartram, Brown, and Conrad (2011) develop the notional- and sensitivity-based measures of hedging intensity and its value and risk consequences that the proxies of Section 6 adopt. What none of these studies has is an exogenous shifter of the *disclosure* environment; they treat hedging as a function of firm fundamentals, whereas this design uses a disclosure mandate as a quasi-experimental shock to the hedging decision itself.

Hedge accounting under IFRS. Glaum and Klöcker (2011) document that hedge-accounting eligibility feeds back into firms’ real hedging choices (“the tail wags the dog”), which is why the hedge-accounting-adoption dummy is a dependent variable here and not merely a control. Panaretou, Shackleton, and Taylor (2013) study the information effects of

IFRS hedge-accounting disclosure in the UK. The designation architecture itself (a combined multi-risk hedge versus split risk components) is treated here as one of the firm’s response margins to disclosure. IAS 39 and IFRS 9 (IASB, 2014) define the effectiveness and cost-of-hedging mechanics.

Mandatory ESG and climate disclosure. The disclosure-economics literature is where the identifying variation lives. Christensen, Hail, and Leuz (2021) survey, on economic-analysis grounds, what mandatory CSR and sustainability reporting can and cannot be expected to do, and caution that real-decision effects (as opposed to reporting effects) are the harder object to identify; the outcome here is accordingly a real choice, hedging. Chen, Hung, and Wang (2018) and Downar et al. (2021) exploit mandated CSR and carbon disclosure to identify real effects; Grewal, Riedl, and Serafeim (2019) study market reactions to the EU nonfinancial-disclosure mandate; Krueger, Sautner, Tang, and Zhang (2024) provide cross-country evidence on the effects of mandatory ESG disclosure. On the climate-risk side, Krueger, Sautner, and Starks (2020) and Ilhan, Krueger, Sautner, and Starks (2023) document that institutional investors price and demand climate-risk disclosure; Bolton and Kacperczyk (2021) document a carbon premium; and Engle, Giglio, Kelly, Lee, and Stroebel (2020) and Hong, Li, and Xu (2019) develop the climate-risk-hedging constructs the climate-hedge ratio h_c operationalizes. What is missing across this literature is a structural link from the disclosure mandate to the *derivatives and hedge-accounting* response specifically; existing work studies emissions, investment, or cost of capital, not the hedge ratio.

Econometrics of staggered adoption. Because Korea’s mandate phases in by tier, the design must use the modern DiD toolkit. Bertrand, Duflo, and Mullainathan (2004) establish the inference problem in DiD with serial correlation; Goodman-Bacon (2021) shows how two-way fixed-effects DiD decomposes, and can be biased, under staggered timing; Callaway and Sant’Anna (2021), Sun and Abraham (2021), and de Chaisemartin and D’Haultfoeulle (2020) supply the heterogeneity-robust group-time estimators this paper uses in place of naive TWFE; and Abadie, Diamond, and Hainmueller (2010) supply the synthetic-control alternative for the largest treated firms, which have the thinnest comparison sets.

3 A Model of Joint Hedging and Disclosure Choice

3.1 Setup and decision variables

A firm carries two zero-mean risk exposures over one period: a financial exposure (the firm’s commodity-and-currency market risk, collapsed for tractability into one factor with volatility σ_f) and a climate-related financial exposure with volatility σ_c , correlated at ρ . The firm chooses three instruments:

$$\begin{aligned} h_f \in [0, 1] \quad (\text{financial hedge ratio}), \quad h_c \in [0, 1] \quad (\text{climate hedge ratio}), \\ d \geq 0 \quad (\text{disclosure intensity}). \end{aligned} \tag{1}$$

Hedging fraction h_f removes that fraction of the financial exposure's standard deviation and h_c that of the climate exposure; the *unhedged* fractions are $1 - h_f$ and $1 - h_c$. Writing the risk covariance matrix

$$\Sigma = \begin{pmatrix} \sigma_f^2 & \rho \sigma_f \sigma_c \\ \rho \sigma_f \sigma_c & \sigma_c^2 \end{pmatrix}, \quad u \equiv 1 - h_f, \quad v \equiv 1 - h_c, \quad \mathbf{u} \equiv (u, v)^\top, \quad (2)$$

the residual exposure (variance of the unhedged portfolio) is the quadratic form

$$R(h_f, h_c) = \mathbf{u}^\top \Sigma \mathbf{u} = (1 - h_f)^2 \sigma_f^2 + (1 - h_c)^2 \sigma_c^2 + 2\rho \sigma_f \sigma_c (1 - h_f)(1 - h_c). \quad (3)$$

A remark on timing. The model is written as one period, but the two choices do not live on the same clock: disclosure intensity is an infrastructure stock (reporting systems, controls, assurance relationships) while hedge ratios are flow decisions reset at each derivative roll. The one-period objective should be read as the per-period value of the corresponding two-stage problem in which d is committed first at amortized cost and the hedges follow each period given d . Because the equilibrium below solves recursively (hedges in closed form given $\Lambda(d)$, then d^* from a scalar fixed point), the static solution coincides with the stationary policy of that two-timescale problem, and nothing in the comparative statics depends on the compression. What the reduction genuinely omits is the *transition*: a mandate arriving mid-stream forces adjustment of the disclosure stock at a pace the model does not price. That omission has an econometric counterpart and not a fix: the event-study coefficients of Section 5 trace the transition path to the new stationary policy, and should not be read as instantaneous jumps between steady states.

3.2 Costs and the firm's problem

Total cost has four components. Hedging carries a linear premium in each leg,

$$C_H = p_f h_f + p_c h_c, \quad (4)$$

with $p_f, p_c > 0$ the per-unit premia (the per-unit cost of the corresponding derivative positions). Disclosure is convex,

$$C_D = a d^2, \quad a > 0, \quad (5)$$

capturing the increasing marginal cost of producing and assuring incremental ESG/climate disclosure. The coefficient a is firm-specific in everything that follows, and no claim is made that it is exogenous in the cross-section: the firm that runs a derivatives desk cheaply plausibly also staffs a reporting function cheaply, so a and risk-management capability are likely entangled. The equilibrium requires only that a be a parameter of the firm's problem; what the *empirical* use of the a -comparative-static requires is the sharper condition that a be predetermined with respect to mandate timing, and the numerical engine (Appendix A) quantifies how a capability confound of this form biases the interaction test, which is why H(sub) is carried as exploratory and not load-bearing. Residual risk carries a baseline economic cost, the distress- and financing-

friction cost of Froot—Scharfstein—Stein,

$$C_R = \phi R(h_f, h_c), \quad \phi \geq 0, \quad (6)$$

and, on top of it, a *disclosure-contingent regulatory penalty*

$$P = \lambda R(h_f, h_c) e^{-kd}, \quad \lambda > 0, \quad k > 0. \quad (7)$$

The penalty is the mechanism. Its magnitude scales with residual exposure R (a firm that carries more unhedged risk is penalized more once that risk is visible), with the regime’s stringency λ (the regulator’s dial), and it is *attenuated* by disclosure: e^{-kd} falls from 1 toward 0 as the firm discloses more, because disclosure converts opaque exposure into quantified, investor-legible exposure that the firm can then be seen to be managing, softening the cost-of-capital and reputational bite of a given R .¹

A screening micro-foundation for the exponential. The functional form need not be defended on tractability alone. Suppose the penalty is levied on residual exposure only while that exposure remains *illegible* to the pricing audience (the regulator, lenders, rating agencies) and that each unit of disclosure intensity generates independent verifiable signals that resolve the firm’s opacity at constant hazard k , the Poisson-verification benchmark of the inspection and auditing literature (Townsend, 1979; Dye, 1986). The probability that the exposure is still unpriced after d units of disclosure is then e^{-kd} , and the *expected* penalty is exactly $\lambda R e^{-kd}$: equation (7) is the reduced form of a verification technology, and k is the informativeness of a unit of disclosure (assurance quality, standardization, machine-readability) and not a free curvature parameter. Two consequences follow. First, the mechanism this foundation pins down is information resolution; a cost-of-capital repricing story generates the same reduced form, and the two are observationally equivalent within it: the model prices the friction without adjudicating its channel (Section 9 returns to this). Second, k acquires an empirical counterpart: disclosure that is assured or filed under a standardized taxonomy should carry a higher k than narrative reporting, a cross-sectional restriction the design can exploit when assurance data become available.

Collecting terms, and defining the *effective marginal price of residual risk*

$$\boxed{\Lambda(d) \equiv \phi + \lambda e^{-kd}} \quad (\Lambda > 0, \quad \Lambda'(d) = -k\lambda e^{-kd} < 0), \quad (8)$$

the firm’s problem is

$$\min_{h_f \in [0,1], h_c \in [0,1], d \geq 0} \Pi(h_f, h_c, d) = p_f h_f + p_c h_c + ad^2 + \Lambda(d) R(h_f, h_c). \quad (9)$$

Equation (9) is the paper’s economic core: disclosure enters *only* through $\Lambda(d)$, so its entire effect on hedging runs through the price the firm puts on residual risk. Raise disclosure and Λ

¹The exponential form is the natural one-parameter attenuation: $P > 0$ for all finite d , $\partial P / \partial d < 0$, and $P \rightarrow 0$ only in the limit of unbounded disclosure, so disclosure never fully substitutes for hedging. It also keeps the disclosure first-order condition analytically closed. A power-law attenuation $R(1 + kd)^{-1}$ yields the same comparative-statics signs.

falls, cheapening residual risk and dampening the hedging motive; raise the regime stringency λ and Λ rises, doing the opposite. The two policy-relevant margins pull hedging in opposite directions, and the equilibrium resolves the tension.

3.3 First-order conditions

Differentiating (3), $\nabla_{(h_f, h_c)} R = -2\Sigma \mathbf{u}$, so with $\mathbf{p} \equiv (p_f, p_c)^\top$ the hedge-ratio first-order conditions are, at an interior optimum,

$$\frac{\partial \Pi}{\partial h_f} = p_f - 2\Lambda(d) [\sigma_f^2(1 - h_f) + \rho\sigma_f\sigma_c(1 - h_c)] = 0, \quad (10)$$

$$\frac{\partial \Pi}{\partial h_c} = p_c - 2\Lambda(d) [\sigma_c^2(1 - h_c) + \rho\sigma_f\sigma_c(1 - h_f)] = 0, \quad (11)$$

which in vector form is the linear system

$$2\Lambda(d) \Sigma \mathbf{u} = \mathbf{p}. \quad (12)$$

The disclosure first-order condition is scalar:

$$\frac{\partial \Pi}{\partial d} = 2ad + \Lambda'(d) R = 2ad - k\lambda e^{-kd} R(h_f, h_c) = 0 \iff \boxed{2ad = k\lambda e^{-kd} R(h_f^*, h_c^*)}. \quad (13)$$

Equation (13) reads cleanly: at the optimum the marginal cost of disclosure, $2ad$, equals the marginal penalty relief it buys, $k\lambda e^{-kd} R$: the rate $k\lambda e^{-kd}$ at which disclosure lowers the penalty, times the residual risk R that the penalty is levied on. Disclosure and hedging are therefore *substitutes in the penalty channel*: a firm that hedges more (smaller R) has less to gain from disclosing, and vice versa.

3.4 Closed-form equilibrium hedge ratios

Because Σ is positive definite (for $|\rho| < 1$, $\sigma_f, \sigma_c > 0$) it is invertible, and (12) solves in closed form:

$$\mathbf{u}^* = (1 - h_f^*, 1 - h_c^*)^\top = \frac{1}{2\Lambda(d^*)} \Sigma^{-1} \mathbf{p}, \quad \Sigma^{-1} = \frac{1}{\sigma_f^2 \sigma_c^2 (1 - \rho^2)} \begin{pmatrix} \sigma_c^2 & -\rho\sigma_f\sigma_c \\ -\rho\sigma_f\sigma_c & \sigma_f^2 \end{pmatrix}. \quad (14)$$

Componentwise, the equilibrium hedge ratios are

$$h_f^* = 1 - \frac{1}{2\Lambda(d^*)(1 - \rho^2)} \left[\frac{p_f}{\sigma_f^2} - \frac{\rho p_c}{\sigma_f\sigma_c} \right], \quad (15)$$

$$h_c^* = 1 - \frac{1}{2\Lambda(d^*)(1 - \rho^2)} \left[\frac{p_c}{\sigma_c^2} - \frac{\rho p_f}{\sigma_f\sigma_c} \right]. \quad (16)$$

Three features are worth stating. First, the hedge vector is the classical minimum-variance solution $\Sigma^{-1} \mathbf{p}$ of Anderson—Danthine and Ederington, scaled by $1/2\Lambda$: the disclosure-and-regulation environment sets the *level* of hedging through Λ , while Σ and $\mathbf{p}$ set its *cross-sectional shape*. Second, full hedging $h = \mathbf{1}$ obtains only in the limit $\Lambda \rightarrow \infty$ (infinite risk price); for

any finite penalty the firm hedges partially, reproducing Stulz’s selective-hedging result as a corollary. Third, the unhedged fraction is proportional to the premium-to-variance ratio net of a correlation credit: a leg is left more open the more expensive its hedge (p) and the smaller its variance (σ^2), the asymmetry commodity-currency hedging programs display on the WTI/FX split, where the low-volatility currency leg is left largely open.

3.5 Equilibrium disclosure and the fixed point

Substituting the optimal hedges into R and using (14),

$$R^* = \mathbf{u}^{*\top} \Sigma \mathbf{u}^* = \frac{1}{4\Lambda(d^*)^2} \mathbf{p}^\top \Sigma^{-1} \mathbf{p} \equiv \frac{\kappa}{4\Lambda(d^*)^2}, \quad \kappa \equiv \mathbf{p}^\top \Sigma^{-1} \mathbf{p} > 0. \quad (17)$$

The disclosure FOC (13) then becomes a single equation in d^* ,

$$2a d^* = k\lambda e^{-kd^*} \cdot \frac{\kappa}{4(\phi + \lambda e^{-kd^*})^2} \iff 8a d^* (\phi + \lambda e^{-kd^*})^2 = k\lambda \kappa e^{-kd^*}. \quad (18)$$

The left side is zero at $d^* = 0$ and strictly increasing and unbounded in d^* ; the right side is positive at $d^* = 0$ and strictly decreasing to 0. A unique interior root $d^* > 0$ therefore exists whenever the penalty regime is active ($\lambda > 0$), by the intermediate-value theorem and strict monotonicity, an existence-and-uniqueness result that holds under the model’s maintained functional forms and does not require solving (18) in closed form. The pair $(\mathbf{u}^*, d^*)$ is the equilibrium: (14) gives hedges as a function of d^* , and (18) closes the loop.

3.6 Mandated disclosure: the floor $\underline{d}$

The problem so far lets the firm choose d freely, but the paper’s subject is *mandatory* disclosure, and a mandate is not a nudge to the interior optimum; it is a constraint. Model a mandate as a floor $d \geq \underline{d} \geq 0$ on the disclosure margin the regulation covers, holding the enforcement dial λ fixed; a disclosure regime is then the pair $(\underline{d}, \lambda)$. The Kuhn—Tucker solution is immediate:

$$d^{**} = \max\{\underline{d}, d_v^*\}, \quad (19)$$

where d_v^* is the voluntary interior root of (18), and the equilibrium splits into two regimes with different economics.

Slack regime ($\underline{d} < d_v^*$): the mandate is inframarginal (the firm was already disclosing past the floor) and the entire equilibrium of Sections 3.4—3.5, including the first-order condition (13), is unchanged. *Binding regime* ($\underline{d} \geq d_v^*$): the floor pins $d^{**} = \underline{d}$, the voluntary FOC becomes the inequality $2a\underline{d} \geq k\lambda e^{-k\underline{d}} R^*$, and hedging is read off (14) at $\Lambda(\underline{d})$, now a mechanical function of the mandated level. Its derivative is signed:

$$\left. \frac{\partial h_f^*}{\partial \underline{d}} \right|_{\text{binding}} = \underbrace{\frac{\partial h_f^*}{\partial \Lambda}}_{>0} \cdot \underbrace{\Lambda'(\underline{d})}_{<0} < 0. \quad (20)$$

A binding, exposure-relevant floor therefore *weakly lowers* hedging: forced disclosure buys the firm penalty relief it did not pay for, cheapens residual risk, and crowds out the hedge. The

two levers of a disclosure regime thus move hedging in *opposite* directions (stringency λ raises it (21), a binding content floor $\underline{d}$ lowers it (20)), so “more disclosure regulation” is not a well-posed treatment until one says which lever moved. H1 is stated for the stringency lever; the crowding-out sign is a distinct, so-far-untested prediction the climate mandate’s intensity margin could eventually identify.

The regime split also disciplines what the executed tests can find. The floors realized in Korea to date (Section 7) constrain governance narrative and physical environmental quantities, content that does not enter the penalty base R or its legibility k on the market-risk margin the model prices. In the model’s terms such a mandate raises $\underline{d}$ on a margin *orthogonal* to Λ : neither λ , k , nor R moves, and the predicted treatment effect on hedging is zero and not merely small. The tightly bounded nulls of Section 7 are therefore the binding-but-orthogonal cell of the $(\underline{d}, \lambda)$ design read back through (19), and the risk-specificity interpretation of Section 7.4 is the verbal statement of this formal one. Finally, the floor clarifies what taking (13) to data requires. Where disclosure is set by regulation and not chosen, the voluntary FOC need not hold in the cross-section; H1—H4 do not condition on it (they are comparative statics of Λ , which is well-defined in both regimes), and only H5 and the d^* -side predictions require the slack regime, which is why Section 4 carries them as conditional.

3.7 Comparative statics

The tested predictions are the signed derivatives of the equilibrium with respect to the policy and structural parameters. I present the direct (partial-equilibrium) signs first (the effect through the explicit channel holding d^* fixed), then the general-equilibrium correction through the disclosure fixed point, and show the correction never overturns the direct sign.

(i) **Stringency λ .** Directly, $\partial\Lambda/\partial\lambda = e^{-kd} > 0$, and from (15),

$$\frac{\partial h_f^*}{\partial\lambda} = \frac{e^{-kd}}{2\Lambda^2(1-\rho^2)} \left[\frac{p_f}{\sigma_f^2} - \frac{\rho p_c}{\sigma_f \sigma_c} \right] = \frac{e^{-kd}}{2\Lambda^2} u_{(\lambda\text{-free})}^* > 0, \quad (21)$$

positive whenever the interior unhedged fraction is positive (the maintained interior condition). Symmetrically $\partial h_c^*/\partial\lambda > 0$. A stricter regime raises hedging on both legs: **H1**, and this sign survives the general-equilibrium feedback through d^* (Appendix A). The disclosure response is not so simple. Totally differentiating the fixed point (18) (Appendix A) gives

$$\text{sign}\left(\frac{\partial d^*}{\partial\lambda}\right) = \text{sign}(\phi - \lambda e^{-kd^*}), \quad (22)$$

a boundary the model earns from the substitutability embodied in (13): a higher λ raises hedging, which *lowers* residual risk R^* , which *weakens* the marginal benefit of disclosure, an effect that opposes the direct penalty-visibility incentive to disclose more. When the firm’s baseline distress-cost price ϕ exceeds the marginal regulatory-penalty price λe^{-kd^*} , the direct incentive wins and disclosure rises with stringency; when the regulatory penalty already dominates, the substitution effect wins and *tougher regulation lowers voluntary disclosure intensity* as the firm shifts protection onto the hedge. “Stricter regime $\Rightarrow$ more disclosure” is therefore a conditional

prediction, and **H2** is accordingly re-anchored (Section 4) to the robust hedging channel and not to d^* .

(ii) Disclosure cost a . Here the channel is entirely indirect and is the model’s most distinctive prediction. From (18), higher a lowers d^* (the left side rises in a , forcing d^* down to restore equality): $\partial d^*/\partial a < 0$. Lower d^* raises e^{-kd^*} , hence raises $\Lambda(d^*)$, hence, through (15), *raises* h_f^* :

$$\frac{\partial h_f^*}{\partial a} = \underbrace{\frac{\partial h_f^*}{\partial \Lambda}}_{>0} \cdot \underbrace{\frac{\partial \Lambda}{\partial d^*}}_{<0} \cdot \underbrace{\frac{\partial d^*}{\partial a}}_{<0} > 0. \quad (23)$$

When disclosure is expensive, the firm discloses less, faces a higher effective residual-risk price, and *substitutes toward hedging*: **H(sub)**. This disclosure-cost-to-hedging substitution is testable through cross-sectional interaction of the treatment effect with proxies for firm-level disclosure cost (assurance fees, reporting complexity, prior reporting infrastructure).

(iii) Correlation ρ . From (15), write $h_f^* = 1 - \frac{1}{2\Lambda}g(\rho)$ with $g(\rho) = \frac{p_f/\sigma_f^2 - \rho p_c/(\sigma_f\sigma_c)}{1 - \rho^2}$. Differentiating,

$$\frac{\partial h_f^*}{\partial \rho} = -\frac{1}{2\Lambda} g'(\rho) = -\frac{1}{2\Lambda} \cdot \frac{2\rho(p_f/\sigma_f^2) - (p_c/\sigma_f\sigma_c)(1 + \rho^2)}{(1 - \rho^2)^2}. \quad (24)$$

The sign is pinned down: $\partial h_f^*/\partial \rho > 0$ iff $2\rho p_f/\sigma_f^2 < (1 + \rho^2)p_c/(\sigma_f\sigma_c)$. In particular at $\rho = 0$, $\partial h_f^*/\partial \rho = \frac{p_c}{2\Lambda\sigma_f\sigma_c} > 0$: starting from independence, more positively correlated exposures raise the optimal financial hedge, because the two risks compound in R and hedging one also relieves the cross term. The prediction reverses only when the own-leg premium-to-variance ratio is large relative to the cross leg’s, a closed-form boundary the empirics can locate and not assume.

(iv) Premia and volatilities. $\partial h_f^*/\partial p_f = -\frac{1}{2\Lambda\sigma_f^2(1-\rho^2)} < 0$ (a costlier hedge is used less) and $\partial h_f^*/\partial \sigma_f > 0$ over the relevant range (a riskier exposure is hedged more), both matching the cross-hedging benchmark.

Table 2 collects the signs and the hypothesis each maps to; Figure 1 draws the equilibrium: the price schedule, the hedge response with its corner and the crowding-out of a binding floor, and the disclosure fixed point.

3.8 Corner solutions

The interior solution requires $h_f^*, h_c^* \in [0, 1]$, i.e. $0 \leq \Sigma^{-1}\mathbf{p} \leq 2\Lambda\mathbf{1}$ componentwise. Two corners matter empirically. If a leg’s premium-to-variance ratio is high enough that (15) returns $h_f^* < 0$, the Kuhn—Tucker solution sets $h_f^* = 0$ (no hedging of that leg) and re-optimizes the other; this is the familiar “leave the cheap-to-carry, expensive-to-hedge leg open” corner that budget-constrained programs produce for FX. If the penalty is severe enough that (15) returns $h_f^* > 1$, the constraint binds at $h_f^* = 1$ (full hedging). The comparative statics of Section 3.7 then apply only in the interior regime; at a binding corner $\partial h_f^*/\partial \lambda = 0$ locally, which is itself a testable non-linearity: the treatment effect should attenuate for firms already at full (or zero) hedging

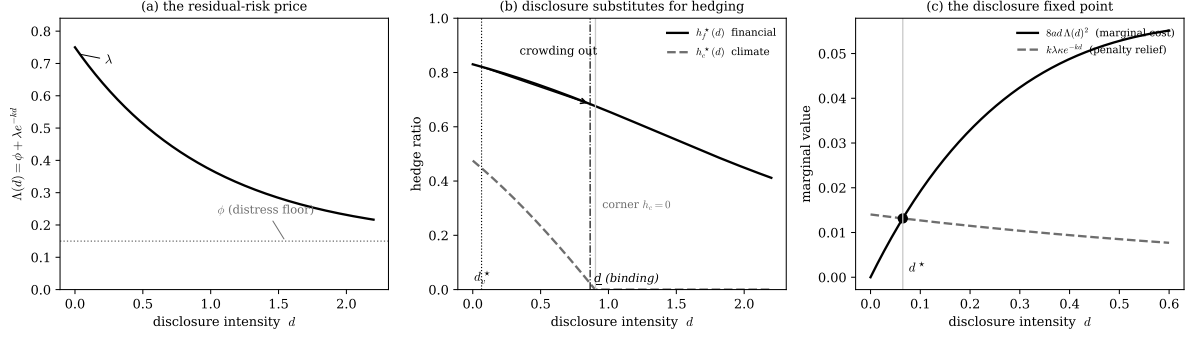


Figure 1: The model in one row, at the calibration of Appendix C. (a) The residual-risk price $\Lambda(d) = \phi + \lambda e^{-kd}$: stringency λ sets the height of the penalty component, disclosure attenuates it toward the distress floor ϕ . (b) Equilibrium hedge ratios as disclosure varies: both legs fall as disclosure cheapens residual risk; the climate leg reaches its $h_c = 0$ corner (Section 3.8); pushing a binding floor $\underline{d}$ past the voluntary optimum d_v^* walks the financial hedge down the same curve, the crowding-out of (20). (c) The disclosure first-order condition (18): marginal cost rising from zero, marginal penalty relief falling, a unique interior crossing at d^* .

pre-mandate. The empirical design accommodates this with a Tobit/two-part specification (Section 5) and not assuming interiority.

A related non-convexity deserves explicit treatment because it is where the linear-premium assumption $C_H = p_f h_f + p_c h_c$ is most vulnerable: real hedging programs carry a fixed participation cost per leg (ISDA documentation, margin and collateral infrastructure, treasury staffing): call it $F_j > 0$. With fixed costs the firm's policy is two-stage: participate in leg j if and only if the interior program's cost saving relative to $h_j = 0$ exceeds F_j , and *conditional on participation* the interior solution (14) is unchanged, since F_j is sunk at the intensive margin. Fixed costs therefore act on the extensive margin only: they generate the $h = 0$ corner for firms whose exposures are too small for the saving (which scales with $\Lambda \sigma^2$) to clear the threshold, predicting non-participation concentrated in small and low-volatility firms. That is the shape of the executed panel, where derivative use is far from universal and strongly size-graded (Section 7); the two-part estimator already separates this participation margin from the intensity margin the interior model governs. Fixed costs would leave the comparative statics intact while disturbing their smoothness at the participation boundary: treatment effects concentrate on firms near the threshold, a heterogeneity the design's size-interaction cells can surface but the baseline ATT averages over.

4 Testable Hypotheses

The comparative statics translate directly into hypotheses stated in terms of observable proxies (defined in Section 6). Let *Treated* index firms brought under the mandatory KSSB/ISSB-aligned disclosure regime and *Post* the periods after their phase-in.

- **H1 (hedging intensity).** Mandatory disclosure raises derivative-based hedging: the DiD coefficient on *HedgeIntensity* (derivative notional/total assets), *FXHedgeRatio*, and *CommodityHedgeRatio* is positive. (From $\partial h_f^*/\partial \lambda > 0$.)

- **H2 (hedge-accounting adoption).** Mandatory disclosure raises the probability of electing IFRS 9 cash-flow-hedge accounting: the DiD coefficient on *HedgeAccountingDummy* is positive. (*Anchored to the robust hedging channel, not to d^* : because H1 raises the hedging level, and the incentive to elect hedge accounting rises with how much a firm hedges (more hedging means more P&L volatility to neutralize through OCI deferral, Glaum—Klöcker’s “tail wags the dog”), adoption inherits H1’s unconditional positive sign, independent of the ambiguous disclosure response (22).*)
- **H3 (carbon-intensity heterogeneity).** The effect is stronger in carbon-intensive industries, where climate exposure σ_c and its correlation ρ with financial exposure are larger: the triple interaction *Post* $\times$ *Treated* $\times$ *CarbonIntensity* is positive. (*From $\partial h^*/\partial \sigma_c > 0$ and (24) at empirically positive ρ .*)
- **H4 (risk-management efficiency).** Firms with higher post-mandate disclosure achieve greater risk reduction per unit of hedging cost (*RiskEfficiency* = $\Delta \text{Vol}/C_H$ rises) because disclosure and hedging are jointly optimized and not either alone. (*From the envelope property of (9): at the joint optimum the marginal risk reduction per KRW of hedging equals $\Lambda(d^*)$, and better-disclosing firms reach a lower- Λ , more cost-efficient point on the same frontier.*)
- **H5 (disclosure-response ordering: making the boundary falsifiable).** The model’s boundary (22), $\text{sign}(\partial d^*/\partial \lambda) = \text{sign}(\phi - \lambda e^{-kd^*})$, is not directly testable because the distress-cost parameter ϕ is latent: without discipline, any observed disclosure response could be rationalized ex post by placing ϕ on the required side of the boundary. Proxying ϕ with *observable* distress-cost measures (leverage, Altman-Z, credit rating, cash-flow volatility) removes that escape hatch and converts the boundary into a signed cross-sectional *ordering*: the post-mandate change in *DisclosureIntensity* should be increasing in distress-cost proxies (positive for high- ϕ firms, non-positive for low- ϕ firms). The interaction *Post* $\times$ *Treated* $\times$ *DistressCost* on disclosure is thus a genuine, rejectable prediction; a null or reversed ordering falsifies the substitution mechanism and not being absorbed by the latent parameter. (*From (22), made empirical.*)
- **H(sub) (disclosure-cost substitution, exploratory).** Among treated firms, those facing higher disclosure cost a raise hedging *more*, not less: *Post* $\times$ *Treated* $\times$ *DisclosureCost* is positive on hedging outcomes while negative on disclosure intensity. (*From (23), the model’s sharpest and least-obvious prediction.*)

H(sub) is the model’s most distinctive prediction but also its least cleanly identified one, and I flag it as exploratory and not confirmatory for a specific reason. The available proxies for disclosure cost a (assurance and audit fees, reporting-complexity indices, the absence of prior sustainability-reporting infrastructure) are not exogenous: they are mechanically entangled with firm size, internal-control quality, and risk-management capacity itself. A firm with weak disclosure infrastructure is disproportionately likely to have weak hedging infrastructure, so the naive interaction confounds the structural substitution channel ($\partial h_f^*/\partial a > 0$) with an omitted-capability channel (weak infrastructure $\Rightarrow$ *less* hedging) that pushes the coefficient the *opposite*

way. The bias is directional: because the capability confound biases the estimate *against* H(sub), a robust positive interaction is conservative evidence for the mechanism, while a null or negative estimate is uninformative: it cannot distinguish “no substitution channel” from “substitution channel masked by capability.”

Identifying H(sub) therefore requires purging the capability content of a : I residualize *DisclosureCost* on size, an internal-control-quality index, and industry before interacting it; restrict comparisons to within size $\times$ industry bins; and, where feasible, instrument disclosure cost with plausibly exogenous variation in assurance requirements (auditor-rotation-driven fee shocks, or the mandatory-assurance phase-in itself). H(sub) is reported with these safeguards and read as suggestive and not load-bearing; that role belongs to H1—H2 under the staggered DiD. A further caveat: the residualization that purges the capability content of a also strips out much of a ’s *structural* variation, since disclosure cost, firm size, and internal-control quality are collinear, so the residualized regressor is thin, and given that the structural effect is second-order to begin with (Section C), H(sub) is likely underpowered even when correctly signed, and is retained as a mechanism probe.

4.1 Predicted estimates and falsification

The design’s predicted results are the model’s comparative-statics signs read onto the DiD coefficients (Table 3), together with the pattern that would *falsify* the mechanism and not merely fail to confirm it. Two features of the mapping matter for execution. First, the outcomes differ sharply in how cleanly they can be measured at scale (Section 6.2): the binary *HedgeAccountingDummy* (H2) is extractable from designation language across the full panel, whereas the continuous hedging-intensity outcomes (H1, H4) require firm-level note adjudication and are therefore estimated on the smaller commodity-exposed subsample. H2 is accordingly the primary, full-panel test; H1/H4 are the corroborating subsample tests, with inference (Section 5.3) chosen to remain valid at small cohort sizes. Nor is the binary outcome a detour from the theory: participation is the extensive margin of Section 3.8, whose fixed-cost threshold moves with Λ (the saving from participating scales with $\Lambda\sigma^2$), so designation and derivative use carry the same signed Λ -prediction as the interior ratios. Second, H5 and H(sub) are heterogeneity tests that survive or fail on their own signed interactions, independent of the average effect.

5 Empirical Design

5.1 Baseline difference-in-differences

The workhorse specification is the two-way fixed-effects DiD

$$Y_{it} = \alpha + \beta (\text{Treated}_i \times \text{Post}_t) + \gamma^\top X_{it} + \mu_i + \tau_t + \varepsilon_{it}, \quad (25)$$

where Y_{it} is a hedging outcome (H1—H4), μ_i are firm fixed effects, τ_t year (or quarter) fixed effects, and X_{it} the standard hedging determinants (size, leverage, market-to-book, foreign-sales share, quick ratio, dividend policy, and cash-flow volatility), following Géczy et al. (1997) and Nance et al. (1993). The coefficient of interest is β , the average treatment effect on the treated

(ATT). Standard errors are clustered by firm; given the serial-correlation concern of Bertrand, Duflo, and Mullainathan (2004), I also report wild-cluster-bootstrap and industry $\times$ year two-way-clustered errors.

5.2 Event study and parallel trends

Identification rests on parallel pre-trends. The event-study form replaces the single interaction with leads and lags relative to firm i 's phase-in year g_i :

$$Y_{it} = \alpha + \sum_{\ell=-L, \ell \neq -1}^{+M} \theta_{\ell} \mathbf{1}\{t - g_i = \ell\} + \gamma^{\top} X_{it} + \mu_i + \tau_t + \varepsilon_{it}, \quad (26)$$

normalizing $\theta_{-1} = 0$. The parallel-trends test is the joint hypothesis $\theta_{\ell} = 0$ for all $\ell < -1$ (a Wald test on the pre-period leads); the dynamic treatment path is read from $\{\theta_{\ell}\}_{\ell \geq 0}$. Confidence bands are plotted with uniform (sup- t) bands, and I report the Rambachan—Roth-style sensitivity of the post-period estimates to bounded violations of parallel trends and not treating a failure-to-reject on the leads as proof.

5.3 Staggered timing and heterogeneity-robust estimation

Because Korea's mandate phases in by listing tier and size, treatment timing g_i varies across firms, and two-way FE (25) is contaminated by “forbidden comparisons” of later-treated to already-treated units (Goodman-Bacon, 2021). The primary estimates therefore use the Callaway—Sant’Anna (2021) group-time ATT $\text{ATT}(g, t)$, aggregated to an overall and to an event-time path, with never-treated (not-yet-covered) firms as the comparison group; I corroborate with the Sun—Abraham (2021) interaction-weighted estimator and the de Chaisemartin—D’Haultfœuille (2020) DID_M . Agreement across these and the naive TWFE is itself a diagnostic: a large gap would signal heterogeneity-driven bias in the latter.

A caveat on statistical power follows directly from the extraction findings of Section 6.2, and it governs how the two tiers of Section 4.1 are estimated. The group-time ATT is only asymptotically valid when each treatment cohort has enough units to estimate $\text{ATT}(g, t)$ precisely; where the outcome requires firm-level note adjudication, as with the continuous hedging-intensity variables (H1, H4) on the commodity-exposed subsample, cohort cell counts are small, and a fully saturated $\text{ATT}(g, t)$ would be noisy. Two responses address this. First, the *primary* test (H2, hedge-accounting adoption) is a designation dummy extractable across the *full* panel, so its cohorts retain the cell sizes the estimator needs; the continuous outcomes are corroborating. Second, on the small subsample the disaggregated group-time ATT is replaced by its *cohort-collapsed* aggregate (a single event-time-weighted ATT and not the full (g, t) grid), with wild-cluster-bootstrap inference designed for few treated clusters (Bertrand, Duflo, and Mullainathan, 2004). The heterogeneity-robust machinery is thus used at the resolution the data support.

5.4 Alternative identification

Four alternatives probe the robustness of β :

1. **Synthetic control** (Abadie et al., 2010) for the largest first-phase treated firms, which have the thinnest natural comparison sets; the donor pool is not-yet-covered firms matched on pre-period hedging and fundamentals.
2. **Triple differences** adding carbon intensity as the third dimension (H3): $Y_{it} = \dots + \beta_3(\text{Treated}_i \times \text{Post}_t \times \text{CarbonIntensity}_i) + \dots$, which differences out any contemporaneous shock common to treated and control within an industry.
3. **Instrumental variables**: instrument firm-level disclosure intensity with the phase-in *eligibility threshold* (asset/market-cap cutoff) interacted with time: a firm’s distance to the regulatory cutoff shifts disclosure obligation but is plausibly excludable from hedging except through the mandate. A regression-discontinuity in the cutoff neighborhood is the local version.
4. **Placebo / falsification**: (a) a pseudo-mandate assigned two years early should yield $\beta \approx 0$; (b) outcomes with no penalty-channel link (e.g. operating-lease intensity) should be unaffected; (c) H(sub)’s reverse-sign prediction (Section 4) is itself a falsification test that a mechanical-quality story fails.

6 Data and Variable Construction

6.1 Sources and reproducible pipeline

The panel is built from seven sources, all machine-retrievable, assembled under a strict reproducibility discipline:

1. **OpenDART** (FSS open API): periodic reports, business reports, and the structured filing index, keyed by corporate registration number (`corp_code`).
2. **XBRL financial statements**: consolidated and separate statements via the OpenDART `fnl` and `ttXbrl` endpoints, parsed on standard K-IFRS tags for total assets, leverage, sales, and foreign sales.
3. **Derivatives footnotes**: the financial-instruments notes (K-IFRS 1107/1109), from which derivative notional amounts by risk type (FX, interest rate, commodity) and fair values are extracted.
4. **Hedge-accounting notes**: the cash-flow-/fair-value-hedge designation disclosures, cash-flow-hedge reserve (OCI) movements, and reclassification amounts.
5. **Sustainability / ESG reports**: KSSB/ISSB-aligned climate disclosures (Scope 1—3 emissions, climate-risk quantification), plus third-party ESG ratings for validation.
6. **KRX market data**: prices, returns, market capitalization, and realized volatility for the risk-reduction outcome.
7. **FnGuide**: analyst, ownership, and standardized-fundamental variables and industry (KSIC) classification.

The pipeline is a deterministic sequence (pull filing index $\rightarrow$ resolve `corp_code` $\times$ period $\rightarrow$ fetch XBRL and notes $\rightarrow$ regex/tag-extract derivative and hedge fields $\rightarrow$ merge KRX/FnGuide $\rightarrow$ winsorize $\rightarrow$ panel), so that every reported number traces to a filing coordinate.

6.2 Executed pilot extraction

The front of this pipeline was executed against the live OpenDART API. The full listed universe was resolved from `corpCode.xml` (118,482 entities; 3,976 with a stock code), and a pilot sample of fourteen firms spanning the four commodity-exposed sectors plus a broad/FX comparison group was pulled for fiscal years 2020—2023 through the `fnl1ttSinglAcntAll` consolidated-statement endpoint. Table 4 reports the resulting real figures. The structured layer behaves as a reproducible denominator source should (total assets, liabilities, and equity return cleanly for every firm-year), but it also reproduces the design’s stated fragility in miniature: the revenue field is null in 11 of 56 firm-years because airlines report it as `yeongeop-suik` and some diversified issuers split it across segment lines that the standard `account_nm` aliases miss, the alias-reconciliation problem Section 6.6 flags. These are real firms at real scale (S-Oil at KRW 21.6tn, POSCO Holdings at KRW 101tn, Samsung Electronics at KRW 456tn total assets in 2023).

The outcome variables the design most cares about (derivative notional/fair value by risk type and the cash-flow-hedge designation) are *not* in the structured account endpoint; they live in the filing notes. That layer was parsed directly: for all fourteen firms the FY2023 annual report (`saeop-bogoseo`) was located through `list.json`, retrieved through `document.xml`, and its derivative note parsed: reading the hedge-accounting designation from the note’s operative sentence, and the derivative figures from the note’s tables. The exercise yields three concrete, measured findings (Table 5, Figure 2).

First, keyword counts mislead; only the operative sentence is reliable. S-Oil’s report contains twelve occurrences of the cash-flow-hedge term, which a keyword classifier would read as adoption, yet the note’s own operative sentence states that “there are no derivatives to which hedge accounting is applied” (*wiheom-hoepi-hoegye ... eops-eumyeo*); the twelve mentions are accounting-policy boilerplate. Reading the designation from that sentence, and not from keyword density, correctly classifies S-Oil as a non-adopter, a classification validated by hand against the filing. Across the pilot the designation was resolvable for ten of fourteen firms (six apply hedge accounting, four do not), the one outcome variable that survives automated extraction because it is read from prose, not from a table.

Second, the numeric derivative tables are not cross-firm comparable. Here the extraction meets a structural wall that no amount of parser effort removes: the derivative note is filed in incompatible shapes across issuers. Of the thirteen firms with a locatable note, exactly *one* (Samsung Electronics) files a single-type, single-unit (KRW) derivative fair-value table that is directly comparable; the other twelve mix table types and units. POSCO Holdings alone files a fair-value table, a notional table, and a profit-and-loss-impact table, in three different units (KRW millions, USD millions, USD thousands); Korean Air’s derivative disclosure is a P&L-impact table (its large figures are period income effects, not fair values); Kia’s blends notional and fair-value columns in one table. A single generic parser cannot return a comparable

“derivative notional” or “fair value” across these without firm-by-firm table adjudication, which is why the notes-level outcomes, unlike the structured denominators, are not a batch job.

Third, where a clean table exists, the numbers are unambiguous. S-Oil’s fair-value sub-table parses to a derivative asset of KRW 3,785 million and liability of KRW 2,149 million, matching the filing line for line, confirming that the parser is correct where the disclosure is clean and that the obstacle is disclosure heterogeneity, not extraction capability.

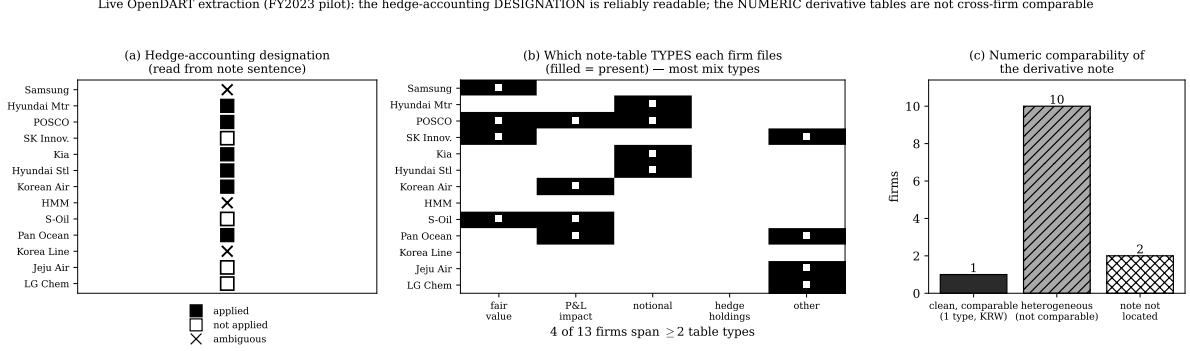


Figure 2: Live OpenDART extraction (FY2023 pilot), in black and white. (a) Hedge-accounting designation read from each firm’s note sentence: applied (filled), not applied (open), ambiguous ($\times$); the one robustly extractable outcome. (b) Which note-table *types* each firm files (filled cell = present); most firms span two or more, so a single parser cannot align them. (c) Numeric comparability of the derivative note: only one of thirteen located notes is a clean single-type table.

6.3 Treatment and sample

The sample is non-financial KOSPI and KOSDAQ firms over a window spanning several years either side of the first mandatory phase-in. Financial firms are excluded (their derivative use is dealing, not hedging). $Treated_i = 1$ if firm i falls in a phase whose obligation has commenced; $Post_t$ is defined relative to each phase’s effective date, so treatment is staggered by g_i . Not-yet-covered firms in later phases are the clean control group.

The FX-hedging and hedge-accounting outcomes (H1 FX leg, H2) run on the full non-financial panel. The *commodity*-hedging outcome, however, runs on a restricted subsample, for a measurement reason. *CommodityHedgeRatio* needs a denominator that reflects genuine physical commodity exposure, and cost of goods sold is a poor universal proxy for it: for a refiner or an airline, COGS *is* essentially the priced commodity (crude, jet fuel), but for a chemical, steel, or auto producer, COGS blends raw materials, labor, depreciation, and semi-finished inputs, so a firm can carry large COGS with little tradable-commodity price exposure, and any commodity-hedge ratio built on it is contaminated. Rather than apply an unreliable proxy panel-wide, the commodity tests are confined to industries whose dominant input is an exchange-traded commodity with an identifiable physical purchase volume (*oil refining, airlines, shipping, and steel*), where the derivative-footnote notional can be scaled by a disclosed physical or clearly commodity-dominated purchase figure and not by blended COGS. This trades sample size for construct validity, the scope chosen for the commodity channel; the FX channel, whose

denominator (foreign sales) is clean across industries, carries the broad-panel evidence.

6.4 Variable definitions

6.5 Mapping the two-risk model to the estimable panel

The model of Section 3 carries two hedge ratios, h_f (financial) and h_c (climate), and consistency requires being explicit about how each reaches the data, and not letting the empirics silently collapse to the financial leg. The financial ratio maps cleanly: h_f is the FX and commodity hedging captured by *FXHedgeRatio*, *CommodityHedgeRatio*, and *HedgeIntensity*, the two-factor WTI/USD-KRW exposure of the motivating importer case. The climate ratio h_c maps to *ClimateHedgeRatio*, the notional of climate-linked instruments (K-ETS emission-allowance derivatives, green and sustainability-linked bonds, carbon-price hedges) scaled by carbon exposure, but this proxy is *sparse*: most Korean non-financial firms did not, over the sample window, hold quantifiable climate-hedging instruments, so h_c is populated only for the disclosing minority (chiefly the carbon-intensive, K-ETS-covered emitters).

This sparsity dictates a two-tier estimation structure. The full-panel tests estimate the financial ratio and the adoption dummy, which is the *financial-risk special case* of the model: formally the ϕ, λ, a mechanism operating on the single factor σ_f , with the climate leg set to zero exposure. The two-risk model is estimated only where it is identified: on the subsample of K-ETS-covered emitters with a non-missing *ClimateHedgeRatio*, where h_c , σ_c , and the cross term $\rho\sigma_f\sigma_c$ are observable and H3’s carbon-intensity heterogeneity has bite. The paper thus nests its own reduction: the general two-risk theory is the object, the financial-risk model is its estimable core on the full panel, and the climate leg is an identified extension on the emitter subsample; no claim about h_c is made on firms for which it is unmeasured.

6.6 Winsorization and missing data

Continuous ratios are winsorized at the 1st and 99th percentiles within each year. Missing derivative-notional fields are treated as structural zeros only when the financial-instruments note affirmatively states no derivative positions; otherwise they are coded missing and multiply imputed for robustness, with results reported both on the imputed and complete-case samples. Firms changing fiscal-year-end or undergoing a merger within the window are flagged and dropped in a robustness cut. The hedge-accounting dummy is validated against the OCI cash-flow-hedge-reserve movement: a non-zero reserve with a zero dummy (or vice versa) is a parsing error and is reconciled by hand.

7 The Executed Study: Korea’s Governance-Report Mandate

The design of Sections 5—6 was written for the KSSB/ISSB climate-disclosure mandate; its implementation plan was finalized in July 2026 with fiscal 2027 as the first covered year (FSC, 2026), so it has no realized post-period yet and cannot yet be tested. Korea does, however, have two ESG-disclosure mandates with realized adoption: the staggered mandatory corporate-governance report (*gieop-jibaegujo bogoseo*), the G pillar, required of KOSPI issuers

with total assets $\geq$ KRW 2 trillion from 2019, $\geq$ 1tn from 2022, and $\geq$ 0.5tn from 2024; and the environmental-information disclosure requirement (Environmental Technology and Industry Support Act), which from 2022 obliges listed firms with assets $\geq$ KRW 2 trillion to report quantitative environmental data. This section estimates the effect of both, with the estimators, outcome parser, and falsification structure of Sections 5—6. Neither mandate quantifies market-risk exposure (the dimension the model prices), so the theory’s prediction for both is a null; the section reports whether the data agree.

7.1 Sample, treatment, and outcomes

From the full OpenDART registry (118,482 entities), 2,391 KOSPI issuers were screened against `company.json` (market classification, industry) and their FY2018 consolidated statements, excluding financials (KSIC 64—66). Cohorts were assigned by FY2018 total assets (an intention-to-treat convention that avoids endogenous threshold-crossing) into $g=2019$ ($\geq$ 2tn; 100 firms), $g=2022$ (1—2tn; 60), $g=2024$ (0.5—1tn; 70), and not-yet-treated ($<$ 0.5tn; 150): 380 firms, with the two upper cohorts close to the population of qualifying KOSPI non-financials. For every firm and fiscal year 2016—2024 the annual report was located, retrieved, and parsed (3,420 firm-years, 3,350 with a filed report) using the designation parser validated in the pilot of Section 6.2. Outcomes: *HedgeAccountingDummy* (1 if the note’s operative sentence affirms hedge-accounting designation; 0 if it affirms non-application *or* the filing mentions no derivative instrument, since designation requires an instrument; missing only where derivatives are present but the designation language is ambiguous; 2,609 usable firm-years across 360 firms), *DerivUse* (any derivative-instrument mention; 3,350 firm-years), and $\ln(1+FXfwd)$, the textual salience of FX-forward disclosure. Treatment $D_{it} = \mathbf{1}\{t \geq g_i\}$.

A data-quality note: DART documents filed before 2021 are encoded CP949 and not UTF-8, and in the initial panel build a decoder default deleted Hangul text from older filings, zeroing all pre-2020 derivative indicators and producing a spurious mandate “effect” of +0.24. The placebo test below flagged the anomaly (a fake mandate on a not-yet-treated cohort also “produced” an effect); correcting the decoder restored economically sensible baselines (a 59% pre-2019 derivative-usage rate) and the estimates reported here. Both panel versions are archived.

Because treatment is assigned by an asset threshold, there is no covariate overlap between treated and control firms, and regression adjustment on size would extrapolate off the support. Comparability is therefore handled through comparison-set design, and every estimate is reported under three rules: (i) all not-yet-treated firms (the Callaway—Sant’Anna default); (ii) the *adjacent* cohort only, the next size tier within roughly a factor of two in assets; and (iii) a *threshold-local* contrast restricted to firms within 50% of each cutoff on either side. A placebo (a fake $g=2019$ mandate assigned to the 2024 cohort over 2016—2023, when its true obligation had not begun) estimates residual size-correlated trends directly.

7.2 Results

Table 7 reports group-time ATTs under all three comparison rules with firm-cluster bootstrap CIs (999 replications), alongside the placebo. Figure 3 shows the event studies with bootstrap bands; Figure 4 displays every estimate against the placebo.

Three findings. *First, the real margins are nulls, robust to the comparison set, with power stated and not implied.* Hedge-accounting adoption moves -0.5pp against a 69% base rate (CI $[-6.8, +6.0]\text{pp}$); derivative usage moves -5.0pp against 61% (CI $[-11.3, +1.3]\text{pp}$); and the signs and magnitudes are stable across all three comparison rules; under the adjacent-cohort rule derivative use is, if anything, significantly *negative* (-0.079 , CI $[-0.156, -0.015]$). Bootstrap standard errors of $3.2\text{--}3.3\text{pp}$ put the minimum detectable effect at 80% power at $9.0\text{--}9.2\text{pp}$, and the operative bound is tighter than that number suggests, because it is one-sided: the confidence intervals cap a *positive* response, the sign the leading alternative channels predict, at $+1.3\text{pp}$ (derivative use) and $+6.0\text{pp}$ (designation). The event-study paths (Figure 3) are flat through event time. *Second, the placebo is clean at this sample size.* Its point estimates ($+0.05$ to $+0.06$ on the real margins) are statistically indistinguishable from zero, so residual size trends are small relative to the reported CIs, and the nulls are a structural feature of the comparison. *Third, the textual margin does not survive scrutiny.* FX-forward disclosure language rises by an insignificant $+0.092$ log points overall; the one marginally significant reading (threshold-local, $+0.158$) is matched in magnitude by the placebo ($+0.102$), consistent with disclosure verbosity trending with size and not responding to the mandate, the reporting-without-real-effects pattern that Christensen, Hail, and Leuz (2021) lead mandatory-CSR research to expect. The carbon-intensity split (H3) yields CIs that include zero in both groups (carbon -0.034 $[-0.171, +0.095]$; non-carbon $+0.009$ $[-0.057, +0.071]$) and is reported as inconclusive.

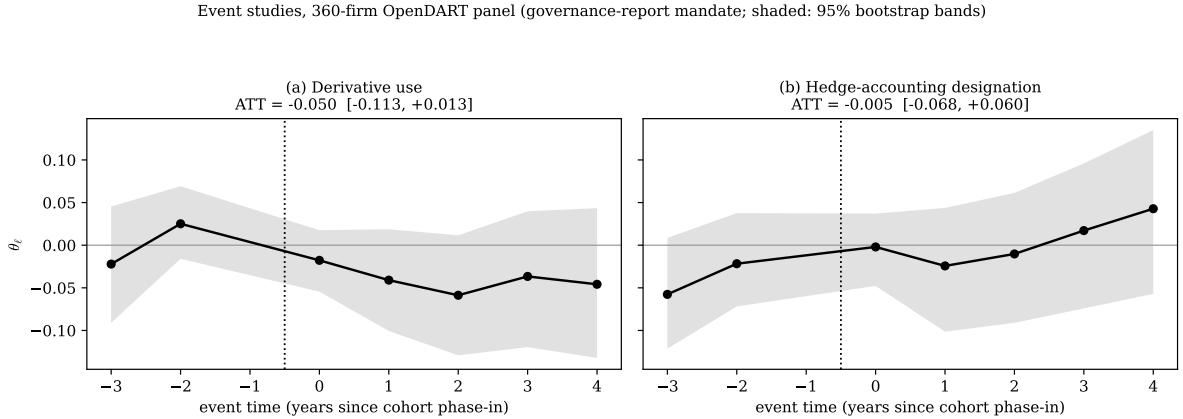


Figure 3: Event studies (380-firm panel; shaded areas are 95% bootstrap bands): (a) derivative use, (b) hedge-accounting designation. Both paths are flat through the phase-in.

7.3 The environmental-information mandate

The second realized mandate provides a sharper test of the same mechanism one rung up the exposure ladder. From 2022, listed firms with assets $\geq$ KRW 2 trillion must disclose quantitative environmental information (emissions, energy, water, waste) under the Environmental Technology and Industry Support Act. This E-pillar mandate forces *quantitative* disclosure, but of physical quantities and not of market-risk exposure; the model therefore still predicts no response in currency and commodity hedging. Estimating the 2022 event on the $\geq 2\text{tn}$ cohort (base year 2021, never-treated controls; the cohort's standing governance obligation is absorbed

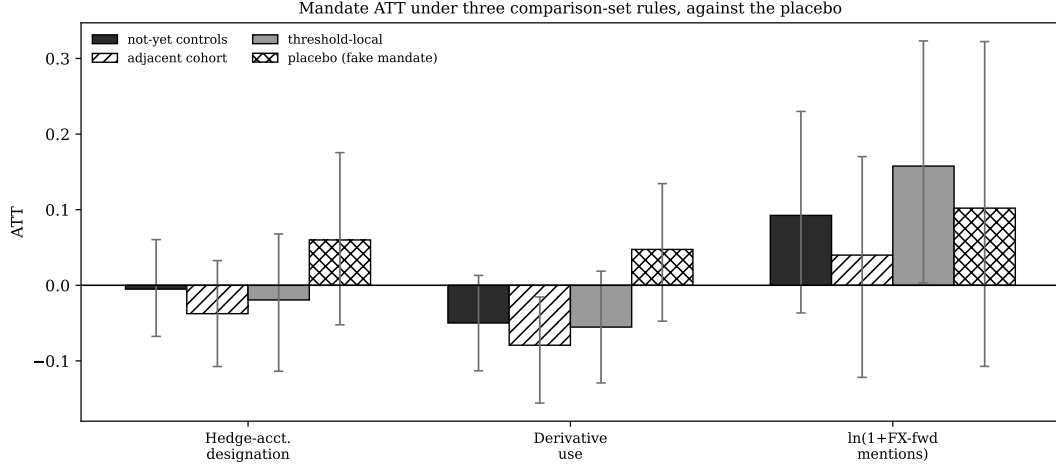


Figure 4: Every governance-mandate ATT under the three comparison-set rules, against the placebo. Real margins cluster at zero across rules; the textual margin’s confidence intervals all cross zero and match the placebo’s magnitude.

by the pre-period) gives ATTs of $+0.010$ $[-0.048, +0.063]$ on hedge-accounting designation and -0.009 $[-0.069, +0.049]$ on derivative use (nulls with half-widths near five percentage points) and $+0.127$ $[-0.038, +0.292]$ on disclosure text. A mandate that compels quantitative environmental reporting moves market-risk hedging no more than the governance report did.

7.4 What the null means for the theory

The model gives the nulls a direct interpretation. In Section 3, disclosure regulation moves hedging only through $\Lambda(d) = \phi + \lambda e^{-kd}$: the price of *residual risk made visible*. The two realized mandates now trace out the bottom of an exposure ladder. A governance report discloses board structure and control without touching exposure at all; a null follows. The environmental-information mandate forces quantitative disclosure, but of physical quantities, not of the market-risk exposure a currency or commodity hedge addresses; a null follows again, and is found, with confidence intervals tight enough to exclude effects beyond roughly six percentage points on either margin.

The nulls discriminate, because the model is not the only theory with a prediction here: a salience channel (a mandate forces the board to confront risk, so risk management changes regardless of content), a governance channel (the report improves oversight, and better-governed firms hedge differently), and a compliance-cost channel (reporting spend crowds out the hedging budget) all predict movement from these mandates. None survives the estimates; the only trace of any of them is the small negative drift in derivative use under the adjacent-cohort rule, which carries the compliance-cost sign and the opposite of the first two. Together the two results say the penalty channel is *risk-specific*: obliging firms to disclose more, even quantitatively, does not change hedging unless what must be disclosed is the hedgeable exposure itself.

Section 3.6 states this formally: both mandates are binding floors $\underline{d}$ on content orthogonal to Λ , the cell of the $(\underline{d}, \lambda)$ design for which the model’s prediction is zero and not merely small. That is what the KSSB/ISSB climate standards will require: S2 compels quantification

of climate-related *financial* exposure, the top rung of the ladder, and the model predicts a hedging response there, negative first and positive after 2030, where the lower rungs produced nothing. If the climate mandate’s realized post-periods deliver the same null on the same pipeline, the penalty-channel mechanism, not one mandate’s scope, is rejected. The pipeline, parser, estimators, and placebo architecture run unchanged on that test.

The finalized climate regime, read through the model. The forward test acquired its dates in July 2026, when the government finalized the implementation plan for KSSB-based sustainability disclosure (FSC, 2026), and each design choice in that plan is legible in the model’s parameters. First, the content floor $\underline{d}$ now sits on the exposure-quantifying margin by statute: climate disclosure under the KSSB standards enters the business report for KOSPI issuers with consolidated assets of KRW 10 trillion or more from fiscal 2027, extending to the 5-trillion tier the following year with a 2-trillion tier under review: a staggered, size-assigned rollout that reproduces the treatment structure the estimators of Section 5.3 are built for. The first wave’s size requires care, and it cuts against the design rather than for it. The figure of roughly 290 entities that appears in the implementation plan counts consolidated subsidiaries as well as filers, which is also why the 2029 figure reaches several thousand; the number of *listed* KOSPI issuers with consolidated assets above KRW 10 trillion, which is the unit this panel observes, is on the order of fifty to sixty. That is a small first cohort, and it consists of the largest issuers in the market, which Section 8 identifies as the firms with the thinnest comparison sets under every one of the three comparison rules. The forward test’s first cohort is therefore its weakest, and power accumulates only as the 5-trillion and 2-trillion tiers phase in.

Second, the final plan moved the *stringency* lever, not only the content floor: the draft’s exchange-level disclosure was upgraded to statutory filing under the Capital Markets Act, with the liability exposure that filing status carries. In the model’s terms this is a rise in λ arriving together with the rise in $\underline{d}$, making this the first Korean mandate to move both levers on the hedging-relevant margin.

That is not, however, enough to sign the response, and the paper does not claim it is. Section 3.6 establishes that the two levers push in *opposite* directions: a rise in stringency raises hedging (21), while a binding content floor lowers it (20). When both move, the sign of the net effect depends on their relative magnitudes, which the model does not pin down. Any prediction of the form “both levers move, hence hedging rises” would be a non-sequitur.

What rescues the forward test is that the finalized plan does not move the two levers at the same time.

Third, the plan phases its own enforcement: liability for disclosure content is largely waived for the first three reporting years, third-party assurance binds from 2030, and Scope 3 quantification is deferred roughly three years. Read through $\Lambda(d) = \phi + \lambda e^{-kd}$, these are a λ that steps up when liability attaches and a k that steps up when assurance makes a unit of disclosure more informative, both on a known calendar.

The separation is what makes the test sharp. Statutory filing begins with fiscal 2027, so $\underline{d}$ binds immediately, while liability for disclosure content is largely waived for the first three

reporting years and assurance binds only from 2030, so λ and k step up later. For fiscal 2027–2029 the model therefore has an unambiguous sign after all, and it is the crowding-out sign: the floor is raised on the hedging-relevant margin while the penalty that would make disclosure a substitute for hedging has not yet attached, so (20) applies alone and hedging should fall. From 2030, when liability and assurance attach, (21) takes over and hedging should rise.

The forward test therefore sharpens from a single post-period sign into a dated, *non-monotone* path prediction: a small negative response over fiscal 2027–2029, followed by a positive response concentrating around 2030–2031. This is a considerably stronger test than a one-sided positive prediction, because it is falsified from three directions rather than one, by a flat path through 2031, by a positive response in the waiver years, and by the absence of a turn when liability attaches. The event-study apparatus of Section 5.2 traces exactly this shape. Figure 5 draws both halves: the executed nulls on the lower rungs against the predicted-positive climate rung, and the regime’s calendar with the predicted response window.

The rejection region, fixed in advance. The forward test’s criteria are set here, on the executed pipeline, before the data exist. The test statistic is the cohort-aggregated group-time ATT on hedge-accounting designation and derivative use for the fiscal-2027 and 2028 cohorts, estimated from OpenDART filings with the parser, comparison rules, and placebo architecture of Section 7.1 run unchanged. The mechanism predicts ATTs that are small through the liability-waiver years and positive with 95% confidence in the window covering fiscal 2030–2031, observed in the 2031–2032 filings. It is rejected if that window delivers a powered null: point estimates near zero with confidence intervals excluding the design’s minimum detectable effect. An underpowered null (intervals covering both zero and the MDE) leaves the test unresolved and will be reported as such, not claimed either way. Two auxiliary escapes, and only two, are admitted in advance: a statutory delay of the regime’s own calendar, or a material dilution of S2’s exposure-quantification content before it binds. Timing is not among them: by fiscal 2031 both the liability step in λ and the assurance step in k have occurred, so a flat path can no longer be attributed to unpriced enforcement.

8 Robustness and Identification Threats

The design’s credibility rests on four identification threats, each with a specific answer. I organize the robustness program around them and not around a list of checks.

Parallel trends. Identification of β requires that treated and control hedging would have moved together absent the mandate. The event study (26) tests this on the pre-period leads with a joint Wald test, sup- t uniform bands, and a Rambachan—Roth-style sensitivity of the post-period estimates to bounded pre-trend violations, so a failure-to-reject on the leads is treated as bounded evidence, not proof. The natural worry that phase order is *endogenous* (the regulator covers the most-exposed firms first) threatens levels, not necessarily trends; the event study addresses trends directly, and the not-yet-covered controls are firms the same regulator will treat later, which tightens comparability. The IV/RD around the size-eligibility cutoff (Section 5.4) exploits that phase membership near the margin is set by the threshold, not by

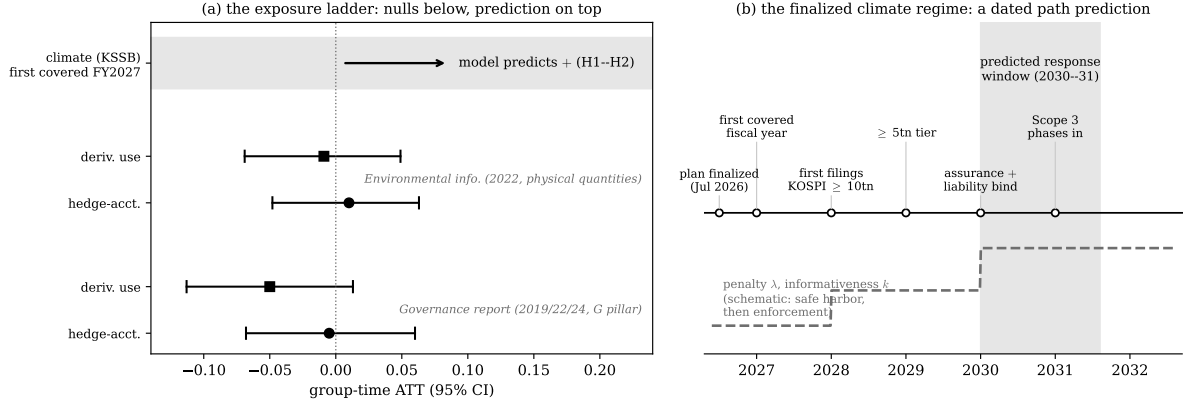


Figure 5: The exposure ladder and the dated forward test. (a) Executed group-time ATTs with 95% bootstrap confidence intervals (not-yet-treated comparison) for the two realized mandates (both nulls on both real margins) beneath the climate rung, where the model predicts a non-monotone response: crowding-out over the liability-waiver years and a positive turn once liability and assurance attach. (b) The finalized climate regime (FSC, 2026) as a calendar: statutory filing begins with fiscal 2027, but liability and mandatory assurance attach from 2030, so the penalty λ and the disclosure informativeness k step up on a known schedule, and the model places the hedging response in the shaded 2030—31 window and not at the first filing.

exposure. A pre-phase-in “treatment” lead separately tests for pre-emptive hedging; a significant lead is reported under both readings, rational anticipation (raising the expected penalty) versus a trends violation.

Measurement error. Three measurement concerns bear on the outcomes. First, derivative notional overstates hedging: a firm can raise notional without reducing risk (Guay and Kothari, 2003); every test is therefore re-run with fair-value- and sensitivity-based (delta-notional) measures where the footnotes permit, and *RiskEfficiency* (H4) is carried so that a notional rise with flat efficiency is reported as an economically distinct (model-consistent corner) outcome and not a confirmation. Second, disclosure text is not hedging substance; this is answered by construction, since the outcomes are real balance-sheet and market objects (derivative notional, OCI cash-flow-hedge-reserve movements, realized volatility), not disclosure language, and boilerplate would attenuate β toward zero, biasing *against* the hypotheses. Third, the commodity denominator (COGS-as-exposure) and the disclosure-cost proxy (capability entanglement) are handled at the construction stage: the former by the commodity-exposed subsample, the latter by residualization and instrumentation (Sections 6.3, 4).

Staggered-adoption bias. Because timing varies by tier, two-way fixed effects (25) is biased by forbidden comparisons of later- to already-treated units (Goodman-Bacon, 2021). The primary estimates are therefore the Callaway—Sant’Anna (2021) group-time ATT against not-yet-covered controls, corroborated by Sun—Abraham (2021) and de Chaisemartin—D’Haultfœuille (2020); TWFE is reported only as a benchmark, and a large TWFE-vs-robust gap is itself read as a heterogeneity diagnostic. Alternative treatment coding (obligation date vs. first affected fiscal year vs. first ISSB-aligned filing) probes sensitivity to the timing definition.

Functional-form dependence. The penalty $\lambda R e^{-kd}$ is a reduced form. The comparative-statics signs of Section 3.7 depend only on $\partial P/\partial d < 0$ and $\partial P/\partial R > 0$, not on the exponential kernel: replacing e^{-kd} with any smooth decreasing attenuation (e.g. the power law $(1 + kd)^{-1}$) preserves every sign, as noted at (7). A structural-estimation extension that recovers (λ, k, a, ϕ) from the data would make the functional form itself testable.

Composition and inference. Entry, exit, and survivorship are addressed with a balanced-panel cut and inverse-probability weighting; overlapping regulations (contemporaneous tax or K-IFRS amendments) are absorbed by year fixed effects and, where firm-specific, by the triple-difference. All headline results carry firm-clustered, industry $\times$ year two-way-clustered, and wild-cluster-bootstrap standard errors, the last for the small number of treated clusters in the first phase (Bertrand, Duflo, and Mullainathan, 2004).

9 Discussion

Three points deserve emphasis. *First, the mechanism operates through a price and not a preference.* Disclosure changes hedging in this model only by changing $\Lambda(d)$, the price the firm puts on residual risk; it does not enter the hedging first-order conditions directly. This is what makes the prediction separable from a general “responsible firms do more” story: within the model, the disclosure-cost substitution result H(sub) has a sign the penalty channel produces and the preference story does not. That separation is a property of the model’s structure, and it identifies the mechanism empirically only to the extent that the maintained assumptions and the identifying variation hold. *Second, disclosure and hedging substitute for one another in risk-bearing while co-moving in observed adoption.* Equation (13) makes them substitutes in the penalty channel (more hedging lowers the disclosure incentive), yet a rise in stringency λ raises *both* in equilibrium (H1 and H2 co-move). A researcher who observes disclosure and hedging rising together and infers complementarity would be conflating the response to λ with the structural relationship at fixed λ , a distinction the triple-difference on disclosure cost is designed to recover. The policy content follows directly: a regulator whose objective is risk-management behavior, and not reporting, should mandate the quantification of the hedgeable exposure itself (positions, notionals, and their ratios to underlying flows) because disclosure content that leaves Λ untouched purchases reports and, on the evidence of Section 7, nothing else. *Third, the model sits one level above the standard hedging toolkit.* Budget-constrained allocation takes p_f, p_c and the covariance matrix and solves for hedge ratios at a *fixed* risk tolerance; this paper endogenizes that tolerance to the disclosure regime. The IFRS 9 designation election (combined versus split), usually studied as an accounting-measurement question, appears here instead as a firm response margin (H2) to the mandate. And the cost of running the hedge is the C_H that the risk-efficiency outcome (H4) normalizes by: one economic object, a firm pricing and covering joint commodity/currency/climate risk, viewed at the decision layer that sets all the others.

Extensions. The design supports four extensions of publishable weight. (1) *The disclosure-quality margin.* Replace the scalar d with a quality-versus-quantity pair and test whether

assured/audited disclosure moves hedging more than volume, separating the penalty channel from a cheap-talk channel. (2) *The cost-of-capital transmission*. Add a mediation layer estimating how much of the hedging response flows through a measured change in cost of debt or equity, linking to Krueger et al. (2020) and Bolton—Kacperczyk (2021). (3) *Real effects beyond derivatives*. Extend the outcome set to operational hedging (input sourcing, currency of debt issuance), testing whether financial and operational hedging are substitutes in the disclosure response. (4) *Structural estimation*. With the closed-form equilibrium of Section 3, the penalty and disclosure-cost parameters (λ, k, a, ϕ) are estimable by matching moments of the hedging and disclosure distributions pre/post mandate, turning the reduced-form DiD into a structural test with counterfactual policy content (e.g. the hedging effect of a stricter future phase).

Limitations. The executed evidence (Section 7) is bounded in scope in three respects. The realized treatments are the governance and environmental-information mandates, not the climate mandate the theory most directly targets, so the nulls constrain the penalty channel’s risk-specificity and not testing its climate form. The outcomes are extensive-margin dummies plus a textual-salience measure, not the continuous hedge ratios of Table 6, whose extraction the pilot showed requires firm-level adjudication. This is a gap between theory and measurement and not merely a data limitation: the model’s objects are the continuous ratios h_f and h_c , and what the executed study observes is whether a firm uses derivatives at all, whether it elects hedge accounting, and how much it writes about currency forwards. H1, which is a statement about hedging *intensity*, has therefore not been tested here; what has been tested is the extensive margin and the designation margin, which the model implies should move in the same direction but whose magnitudes it does not tie to h_f . Readers should treat the reported nulls as nulls on those margins; effects on hedging *intensity* within the always-hedging majority are therefore outside what these estimates bound. And treatment is assigned by an asset threshold with no covariate overlap, so identification rests on comparison-set design and not covariate adjustment: the three comparison rules and the placebo (which is indistinguishable from zero at this sample size) are the response, but a residual size-trend component within the reported CIs cannot be excluded.

A further set of limitations attaches to the ESG disclosure data itself, and bears directly on how much confirmation the structural model can draw from agreement with the panel. ESG disclosure is a self-reported, largely unassured object, and the gap between reported and substantive environmental conduct, commonly discussed as greenwashing, is a documented feature of these filings and not an occasional defect. The disclosure intensity d of Section 3 is a choice variable with a real cost $C_D(d)$ and a real informational payoff; the measured counterpart is a filing whose length, boilerplate share, and assurance status vary widely across issuers and which a firm may increase without changing what it discloses in substance. Measurement error of this kind is unlikely to be classical: firms with the weakest underlying performance have the strongest incentive to inflate disclosure volume, so the error is plausibly correlated with the latent variable it proxies for and with the outcome, and the resulting bias in any d -side coefficient has no sign guaranteed in advance. The same concern applies on the hedging side, where derivative notional proxies for a risk-reduction quantity the model treats as observable.

Endogeneity is the second concern, and it is not fully resolved by the design. Disclosure is chosen jointly with hedging in the model, which is precisely why the empirical strategy relies on regulatory floors and not on cross-sectional variation in d ; but the mandates themselves were legislated in a period when climate risk was rising in salience for the same firms whose hedging behavior is the outcome, so a coincident shift in hedging demand is a live alternative explanation that the triple-difference and placebo-timing tests bound and not eliminate. Reverse causality is a further possibility on the disclosure margin: firms that already hedge extensively may find quantified risk disclosure cheaper to produce, which would generate an association between d and hedging with causality running opposite to the model's. And unobserved firm characteristics, most obviously governance quality, managerial risk aversion, and the sophistication of the treasury function, plausibly drive disclosure and hedging simultaneously; firm fixed effects absorb the time-invariant component of such factors but not their evolution, leaving an omitted-variable channel that the size-threshold assignment does not close.

These considerations condition how the paper's principal structural result should be read. The comparative statics of Section 3.7 follow from the implicit function theorem applied to the model's first-order conditions, and their signs are therefore consequences of the assumed functional forms and not empirical findings. That the Korean nulls of Section 7 are consistent with the model's prediction for a floor raised on a margin orthogonal to Λ is supportive, but a null result is weak evidence for any particular structure taken on its own: many models, including ones with no penalty channel at all, predict no hedging response to a governance-report mandate. The contribution the executed study makes is therefore not the null itself but the *ladder* it sits on. The penalty channel is the mechanism that orders mandates by how directly they quantify hedgeable exposure, and that ordering generates a testable sequence rather than a single coefficient: nulls where the mandate leaves Λ untouched, and a signed, dated, non-monotone path where it does not. The governance and environmental-information rungs establish the lower end of that ladder on an executed pipeline with a placebo indistinguishable from zero; the climate rung, with its rejection region fixed in advance in Section 7.4, is what the design is for. A framework that predicted nothing on the lower rungs would be observationally equivalent here, but it would not also have to commit to the shape of the upper rung. Consistency between a structural prediction and an estimated coefficient does not establish that the mechanism generating the data is the one the model posits, particularly where the outcome is a null and the data carry the measurement and endogeneity problems described above. The forward test on the climate mandate, over which the model predicts a signed non-monotone path rather than merely a non-zero response, is the more informative one for this reason, and the interpretation offered here should be treated as provisional until that test is run.

The theory collapses the firm's multi-factor commodity/currency exposure into a single financial factor σ_f to keep the equilibrium closed-form (a full two-factor treatment extends the model at the cost of a 3×3 system), and the penalty $\lambda R e^{-kd}$, while given a screening foundation in Section 3.2 that pins the exponential to a verification hazard, remains observationally equivalent to a cost-of-capital repricing channel the model does not separate. The remaining theory-to-panel gaps, the sparsity of the climate-hedge ratio and the latency of the distress parameter ϕ , are handled where they arise (Sections 6.5 and 4).

The identifying variation is a single country’s phased mandates, so external validity beyond ISSB-aligned regimes is untested; derivative-notional measurement is coarse and hedge-accounting election is observed only when disclosed. The triple-difference and placebo-timing tests of Section 5 are constructed specifically to separate the mandate’s effect on hedging from a coincident shift in hedging demand driven by the same climate developments that motivated the mandate.

Executing this design on real filings is principally a data-construction task and not a theoretical or econometric one, and the executed pilot of Section 6.2 now locates that task precisely. The closed-form solver, the comparative-statics signs, and the staggered-DiD estimator are verified against model-generated panels (Section C); the identification strategy is specified to the coefficient; and the structured front of the pipeline was run live against OpenDART, returning clean financial fields for a real fourteen-firm, four-year sample. Parsing the derivative notes for those firms, however, surfaces a structural obstacle that is not a matter of parser effort: the hedge-accounting *designation* is reliably readable from the note’s operative sentence (resolved for ten of fourteen pilot firms, and validated by hand for S-Oil), but the *numeric* derivative tables are filed in incompatible types and units across issuers (fair-value, notional, and profit-and-loss-impact tables, in KRW and USD, sometimes several within one firm), so that only one of thirteen located notes is directly comparable (Section 6.2). The implication for the empirical program is concrete: the hedge-accounting-adoption outcome (H2) is extractable at scale, but the continuous hedging-intensity outcomes (H1, H4) require firm-level table adjudication or a supervised extraction model and not a uniform parse. That the binding constraint is disclosure heterogeneity and construct validity, not the economics or the econometrics, is where execution of the full design succeeds or fails.

10 Conclusion

Whether mandatory ESG disclosure changes corporate hedging is, at bottom, a question about the price a firm puts on the risk it chooses not to hedge. This paper makes that intuition precise and then takes it to data. In the model, disclosure enters only through the effective residual-risk price $\Lambda(d) = \phi + \lambda e^{-kd}$; the interior optimum in the two hedge ratios is the closed-form minimum-variance vector $\mathbf{1} - \frac{1}{2\Lambda(d^*)}\Sigma^{-1}\mathbf{p}$; hedging rises unconditionally with stringency while the disclosure response carries the sign of $\phi - \lambda e^{-kd^*}$; and costlier disclosure raises hedging through substitution ($\partial h_f / \partial a > 0$).

The executed test runs the paper’s staggered, heterogeneity-robust design on Korea’s two ESG-disclosure mandates with realized adoption, the governance report (2019/2022/2024 by asset tier) and the environmental-information requirement (2022, assets ≥ 2 tn), on a 380-firm, 3,420 firm-year panel built from OpenDART filings. The answer is a disciplined null, twice: hedge-accounting adoption at -0.005 $[-0.068, +0.060]$ and derivative use at -0.050 $[-0.113, +0.013]$ under the governance mandate, $+0.010$ $[-0.048, +0.063]$ and -0.009 $[-0.069, +0.049]$ under the environmental mandate, stable across three comparison-set rules, with flat event studies, placebos indistinguishable from zero, and no robust movement even in disclosure text.

The nulls are the theory's own reading made empirical: neither mandate forces market-risk exposure into the open, so neither moves the hedging-relevant Λ (the penalty channel is risk-specific), and the KSSB climate mandate, which compels quantification of the exposure itself, stands as this paper's falsifiable forward test: under the implementation plan finalized in July 2026 its first covered fiscal year is 2027, and the plan's phased enforcement (statutory liability and mandatory assurance binding from 2030) converts the test into a dated path prediction and not a single sign, with its rejection region fixed in advance (Section 7.4). The paper's answer to its title question is therefore a mechanism with evidence behind it: what makes the firm want to hedge is disclosure that prices the risk it would otherwise carry quietly, and not the obligation to disclose as such.

A Comparative Statics via the Implicit Function Theorem

The general-equilibrium signs of Section 3.7 follow from totally differentiating the equilibrium system $\{(12), (13)\}$. Stack the equilibrium conditions as $F(\mathbf{u}, d; \theta) = 0$ for parameter θ :

$$F_1 : 2\Lambda(d)\Sigma\mathbf{u} - \mathbf{p} = 0, \quad F_2 : 2ad - k\lambda e^{-kd} \mathbf{u}^\top \Sigma \mathbf{u} = 0. \quad (27)$$

The Jacobian with respect to $(\mathbf{u}, d)$ is

$$J = \begin{pmatrix} 2\Lambda\Sigma & 2\Lambda'(d)\Sigma\mathbf{u} \\ -2k\lambda e^{-kd}(\Sigma\mathbf{u})^\top & 2a + k^2\lambda e^{-kd} \mathbf{u}^\top \Sigma \mathbf{u} \end{pmatrix}, \quad (28)$$

whose determinant is positive under the second-order condition (32) (it is the same Schur complement, up to the positive factor $\det(2\Lambda\Sigma)$). For $\theta = \lambda$,

$$-\frac{\partial F}{\partial \lambda} = \begin{pmatrix} -2e^{-kd}\Sigma\mathbf{u} \\ k e^{-kd} \mathbf{u}^\top \Sigma \mathbf{u} \end{pmatrix}, \quad (29)$$

and Cramer's rule on $J(\partial\mathbf{u}/\partial\lambda, \partial d/\partial\lambda)^\top = -\partial F/\partial\lambda$ delivers $\partial\mathbf{u}^*/\partial\lambda < 0$ componentwise (the unhedged fraction falls), hence $\partial h_f^*/\partial\lambda > 0$, confirming H1 inclusive of the disclosure feedback. The disclosure derivative is not unconditionally signed. Working the scalar fixed point (18) directly: take logs, $\ln(8a) + \ln d + 2\ln\Lambda = \ln(k\lambda\kappa) - kd$, and differentiate totally in λ , which collects to

$$\frac{\partial d^*}{\partial \lambda} \left[\frac{1}{d^*} + k - \frac{2k\lambda e^{-kd^*}}{\Lambda} \right] = \frac{1}{\lambda} - \frac{2e^{-kd^*}}{\Lambda}. \quad (30)$$

The bracket on the left is positive at an interior optimum, so the sign of $\partial d^*/\partial\lambda$ is the sign of the right-hand side, $\frac{1}{\lambda} - \frac{2e^{-kd^*}}{\Lambda}$, which is positive iff $\Lambda > 2\lambda e^{-kd^*}$, i.e. iff $\phi > \lambda e^{-kd^*}$, which is the boundary (22). The same construction with $\theta = a$ gives $\partial d^*/\partial a < 0$ and $\partial\mathbf{u}^*/\partial a < 0$, hence $\partial h_f^*/\partial a > 0$, establishing the structural (level) form of H(sub).

B Second-Order Conditions, Existence, and Uniqueness

The Hessian of Π blocks as

$$H = \begin{pmatrix} 2\Lambda\Sigma & \Lambda'(d)\nabla_h R \\ \Lambda'(d)\nabla_h R^\top & 2a + \Lambda''(d)R \end{pmatrix}, \quad \nabla_h R = -2\Sigma\mathbf{u}, \quad \Lambda''(d) = k^2\lambda e^{-kd} > 0. \quad (31)$$

The hedge block $2\Lambda\Sigma$ is positive definite (Section 3.4). The disclosure diagonal $2a + \Lambda''R = 2a + k^2\lambda e^{-kd}R > 0$. By the Schur-complement criterion, $H \succ 0$ (so the interior critical point is the unique global minimum) iff

$$2a + k^2\lambda e^{-kd}R > \Lambda'(d)^2 \nabla_h R^\top (2\Lambda\Sigma)^{-1} \nabla_h R = \frac{k^2\lambda^2 e^{-2kd}}{2\Lambda} \cdot 2\mathbf{u}^\top \Sigma \mathbf{u} = \frac{k^2\lambda^2 e^{-2kd}R}{\Lambda}, \quad (32)$$

using $\nabla_h R^\top \Sigma^{-1} \nabla_h R = 4\mathbf{u}^\top \Sigma \mathbf{u} = 4R$. Condition (32) rearranges to a lower bound on the disclosure-cost curvature,

$$a > \frac{k^2\lambda e^{-kd}R}{2} \left(\frac{\lambda e^{-kd}}{\Lambda} - 1 \right) = -\frac{k^2\lambda e^{-kd}R}{2} \cdot \frac{\phi}{\Lambda} \leq 0, \quad (33)$$

which is *automatically satisfied* whenever $\phi \geq 0$ and $a > 0$: the right-hand side is non-positive, so any positive disclosure-cost curvature guarantees a strict global minimum. The equilibrium of Sections 3.4–3.5 is therefore well-defined and unique within this parameterization, without further parameter conditions; uniqueness is a property of the assumed quadratic-cost, exponential-attenuation specification and not a general feature of joint disclosure-hedging problems: the substitutability between disclosure and hedging (the negative cross term) is what makes the joint problem convex here.

C Numerical Validation of the Model and the Estimators

Before any real filing is parsed, two things are checked: that the equilibrium of Section 3 is what the closed form claims, and that the estimators of Section 5 recover the predicted signs when the data are generated by the model itself. Both checks run on model-generated data and are not evidence about real firms; they check the machinery on cases with a known answer.

C.1 Structural solver and comparative-statics verification

The firm's program (9) is solved two independent ways, the closed-form fixed point of Sections 3.4–3.5 and a direct multi-start box-constrained minimization of Π , and the two are required to agree. At the baseline calibration ($\sigma_f=0.30$, $\sigma_c=0.15$, $\rho=0.20$, $p_f=0.030$, $p_c=0.020$, $\phi=0.15$, $\lambda=0.60$, $k=1.0$, $a=0.05$) they agree to 8.8×10^{-9} , giving $h_f^* = 0.8213$, $h_c^* = 0.4476$, $d^* = 0.0648$, $R^* = 1.15 \times 10^{-2}$. Every comparative-statics sign of Table 2 is then verified by resolving the full equilibrium under parameter shocks (Table 8). The delicate one is the disclosure response: the measured $\partial d^*/\partial \lambda = -0.065$ is *negative* here, and it should be: at this calibration $\phi - \lambda e^{-kd^*} = -0.412 < 0$, so the boundary (22) predicts the negative sign the solver returns. The substitution channel $\partial h_f^*/\partial a = +0.190 > 0$ (with $\partial d^*/\partial a = -1.346 < 0$) is confirmed as a

genuine but modest level effect. Figure 6 shows the verified curves. The solver also prices the policy experiment the forward test contemplates: stepping the penalty λ from its baseline 0.60 to 0.75, 0.90, and 1.20 (a 21—83% rise in the equilibrium price $\Lambda(d^*)$) moves the financial hedge ratio by +3.1, +5.2, and +8.1pp and the climate hedge ratio by +9.4, +16.1, and +25.0pp, while equilibrium disclosure gives ground (d^* falls from 0.065 to 0.056, 0.049, and 0.040). The response the 2030—31 window should reveal is therefore single-digit on the financial margin, several times larger on the climate margin, and accompanied by substitution out of voluntary disclosure.

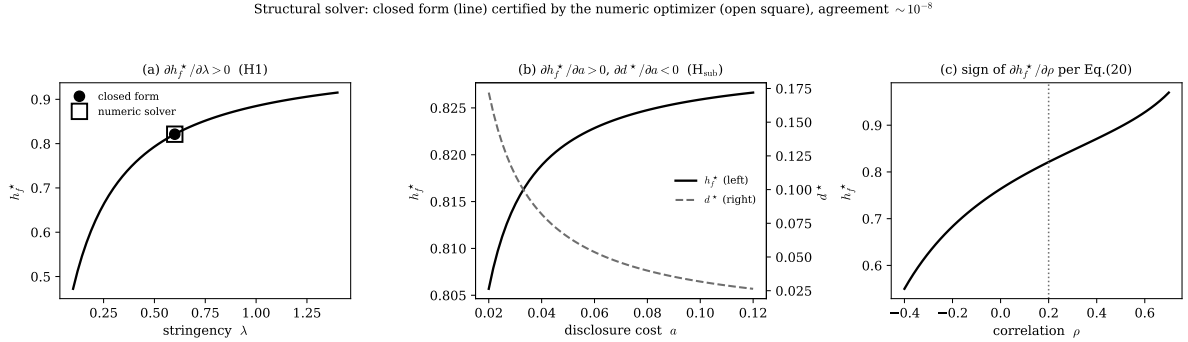


Figure 6: Structural solver. Closed-form equilibrium (line) checked against the independent numeric optimizer ($\times$, panel a). (a) h_f^* rising in stringency λ (H1). (b) h_f^* rising and d^* falling in disclosure cost a : the substitution channel H(sub). (c) h_f^* rising in correlation ρ over the calibrated range, the sign given by Eq. (24).

C.2 Recovery on model-generated staggered panels

A synthetic panel ($N=900$ firms, $T=10$ periods, cohorts phasing in at $g \in \{4, 6, \infty\}$) is generated from the structural DGP: the mandate raises λ for treated firms, with a larger jump for carbon-intensive firms (the H3 channel), and the hedging response phases in over three years, so treatment effects are dynamic and, with staggered timing, two-way fixed effects is expected to be biased. Table 9 reports the result. The Callaway—Sant’Anna group-time ATT recovers the true average effect exactly (0.0622 vs. 0.0621), while TWFE understates it by 12% (0.0549), the textbook heterogeneity-with-dynamics bias; pre-trend coefficients are two orders of magnitude below the effect ($\max |\theta_{\ell < 0}| = 2.9 \times 10^{-4}$ vs. ATT 6.2×10^{-2}). Hedge-accounting adoption (H2, anchored to the hedging level) and the carbon-intensity heterogeneity (H3) both come through with the predicted signs. Figure 7 shows the event-study path (flat before, ramping after) and the estimator comparison.

C.3 H(sub) is the fragile prediction

The engine tests the identification worry of Section 4 directly. On a cross-section of treated firms whose disclosure cost a is *exogenous*, the slope of hedging on a is +0.139: the structural $\partial h_f^* / \partial a > 0$ made visible. Introducing a capability confound of plausible size, letting the same latent (lack of) sophistication that raises disclosure cost also depress risk-management capability, drives the naive slope to -1.708 , flipping its sign entirely; controlling for the capability index

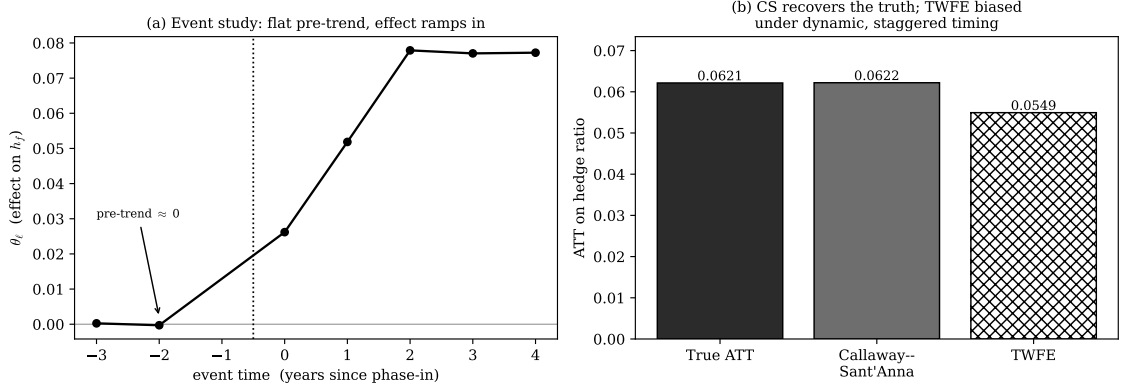


Figure 7: Design recovery. (a) Event study on model-generated data: flat pre-trend, effect ramping in over three post-treatment years. (b) The Callaway—Sant’Anna ATT recovers the true model ATT; TWFE understates it under heterogeneous, dynamic timing.

restores it to $+0.170$ (Figure 8). This is $H(\text{sub})$ ’s fragility made quantitative: the substitution effect is real but second-order, so it is the prediction a realistic confound can overturn, and only a design that purges the capability content of the disclosure-cost proxy (Section 4) can identify it. The engine thus validates both the confirmatory predictions and the boundary condition under which the exploratory one holds.

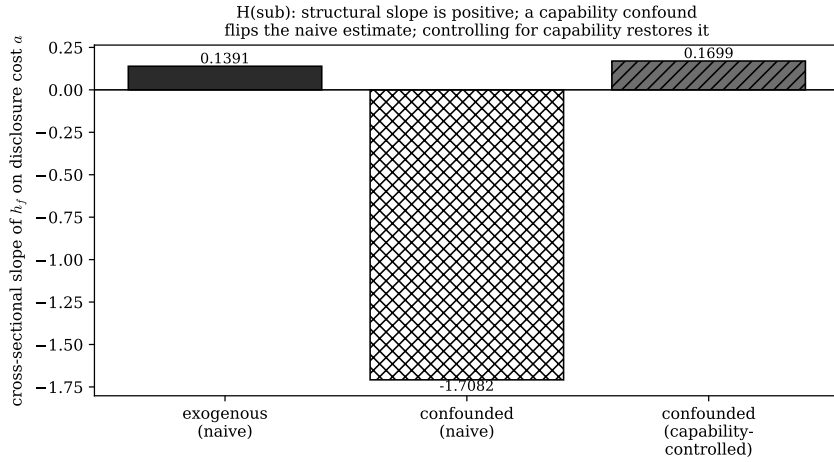


Figure 8: $H(\text{sub})$ fragility. Cross-sectional slope of treated-firm hedging on disclosure cost a : positive when a is exogenous (structural $\partial h_f^*/\partial a > 0$), flipped negative by a capability confound, restored positive by controlling for capability. The substitution channel is real but second-order, hence identifiable only with the safeguards of Section 4.

Category	Representative works	Method / data	How this paper differs
Hedging theory	Smith—Stulz (1985); Froot—Scharfstein—Stein (1993); Stulz (1996)	Analytical models of costly hedging	Lets the residual-risk price be endogenous to a disclosure choice
Cross-hedging	Anderson—Danthine (1981); Ederington (1979); Adler—Dumas (1984)	Minimum-variance ratios	Two coupled exposures with a disclosure-scaled risk price
Derivative use & value	Nance et al. (1993); Géczy et al. (1997); Allayannis—Weston (2001); Bartram et al. (2011); Jin—Jorion (2006)	Cross-sectional / panel; notional & sensitivity proxies	Uses a disclosure mandate as a quasi-experiment on hedging
Hedge accounting	Glaum—Klöcker (2011); Panaretou et al. (2013); IASB (2014)	Survey / archival; IFRS 9 mechanics	Hedge-accounting adoption as an outcome of disclosure regulation
ESG / climate disclosure	Christensen et al. (2021); Chen et al. (2018); Downar et al. (2021); Krueger et al. (2020, 2024); Ilhan et al. (2023)	Mandate-based DiD; investor surveys	Links the mandate to derivatives & hedge accounting, not emissions
Climate-risk hedging	Engle et al. (2020); Hong et al. (2019); Bolton—Kacperczyk (2021)	Return-based hedge portfolios	Operationalizes a firm-level climate-hedge ratio h_c
Staggered DiD	Goodman-Bacon (2021); Callaway—Sant’Anna (2021); Sun—Abraham (2021); de Chaisemartin—D’Haultf. (2020); Abadie et al. (2010)	Group-time ATT; synthetic control	Applies heterogeneity-robust estimators to a phased Korean mandate

Table 1: Literature map. Indispensable works are those defining the residual-risk-price mechanism (Froot—Scharfstein—Stein; Stulz), the hedging-measurement apparatus (Géczy et al.; Bartram et al.), the mandate-as-quasi-experiment template (Christensen et al.; Downar et al.), and the heterogeneity-robust estimator (Callaway—Sant’Anna; Sun—Abraham). The climate-return-hedging works (Engle et al.; Hong et al.) are supporting and not indispensable.

Parameter	Effect on h_f^*	Effect on d^*	Empirical counterpart
Stringency λ	+	$\text{sign}(\phi - \lambda e^{-kd^*})$	Treatment (Post $\times$ Treated): H1, H2
Disclosure cost a	+	—	Disclosure-cost interaction: H(sub)
Correlation ρ	sign per (24)	ambiguous	Carbon-intensity / exposure interaction: H3
Hedge premium p_f	—	+	Cost-of-hedging control
Exposure vol σ_f	+	+	Pre-period exposure control

Table 2: Comparative statics of the equilibrium and their empirical counterparts. Signs hold at an interior optimum; corner cases are treated in Section 3.8.

Hypothesis	Coefficient	Predicted sign	Falsifying pattern
H2 (primary)	β on hedge-acct. dummy (full panel)	+	$\beta \leq 0$, or positive pre-trend leads
H1	β on hedging intensity (subsample)	+	$\beta \leq 0$
H3	β_3 on carbon triple-diff	+	$\beta_3 \leq 0$ (effect not where σ_c, ρ largest)
H4	β on RiskEfficiency	+	efficiency flat or falls (hedging rises but wastefully)
H5	β_3 on distress-cost triple-diff (d outcome)	+	flat or reversed ordering (boundary (22) rejected)
H(sub)	β_3 on disclosure-cost triple-diff	+ on h , — on d	+ on both (quality story, not penalty channel)

Table 3: Predicted estimates and falsification patterns. H2 is the load-bearing, full-panel confirmatory test (the adoption dummy is extractable at scale); H1/H4 corroborate on the adjudicated subsample. H5 makes the disclosure-response boundary (22) falsifiable by proxying the latent ϕ . H(sub) is exploratory and likely underpowered (Section 4).

Sector (pilot firms)	Firm-yrs	Assets (tn KRW)	Leverage	Rev. miss.
Refining (S-Oil, SK Innovation)	8	38.96	0.612	0
Airline (Korean Air, Jeju Air)	8	14.67	0.787	4
Shipping (HMM, Pan Ocean, Korea Line)	12	10.16	0.485	0
Steel (POSCO, Hyundai Steel, KG Steel)	12	43.76	0.465	1
Broad/FX (Samsung, Hyundai, LG, Kia)	16	208.86	0.463	6

Table 4: Executed OpenDART pull (`fnl1ttSinglAcntA11`, consolidated, FY2020—2023). Real retrieved figures. Leverage is total liabilities/total assets. The “revenue missing” column is the count of firm-years where the standard account aliases failed to resolve a revenue line: the alias-reconciliation cost, quantified on live data and not asserted.

Firm (FY2023)	Hedge-acct. designation	Note table type(s)	Numeric comparability
Samsung Electronics	ambiguous	FV (KRW)	comparable (single type)
S-Oil	not applied (validated)	FV + P&L	heterogeneous
POSCO Holdings	applied	FV + notional + P&L (KRW/USD)	heterogeneous
Korean Air	applied	P&L impact	heterogeneous
Kia	applied	notional + FV mixed	heterogeneous
SK Innovation	not applied	FV + other	heterogeneous
HMM	unresolved	note not located	n/a

Table 5: Executed derivative-note parse (`document.xml`); FV = fair value. The hedge-accounting designation is read from the note’s operative sentence and is resolvable for most firms; the numeric derivative tables are filed in incompatible types and units and are not cross-firm comparable without firm-level adjudication. S-Oil’s fair-value figures were validated by hand against the filing. Representative firms shown; totals over the full fourteen-firm pilot: designation resolved 10/14 (6 applied, 4 not applied); numeric notes comparable 1/13, heterogeneous 10/13, not located 2/13; one document non-retrievable.

Variable	Definition	Units / type
<i>HedgeIntensity</i>	Total derivative notional / total assets	ratio, winsor. 1/99
<i>FXHedgeRatio</i>	FX-derivative notional / foreign sales	ratio, winsor.
<i>CommodityHedgeRatio</i>	Commodity-derivative notional / physical (or commodity-dominated) purchases; commodity-exposed subsample only (refining, airlines, shipping, steel)	ratio, winsor.
<i>ClimateHedgeRatio</i> h_c	Notional of climate-linked instruments (K-ETS emission-allowance derivatives, green/sustainability-linked bond issuance, carbon-price hedges) / carbon exposure (Scope 1+2 $\times$ allowance price); disclosing firms only	ratio, sparse
<i>HedgeAccountingDummy</i>	1 if cash-flow-hedge accounting elected (K-IFRS 1109)	binary
<i>RiskEfficiency</i>	Realized-volatility reduction / hedging cost ($\Delta \text{Vol}/C_H$)	ratio, winsor.
<i>DisclosureIntensity</i> d	Standardized index of KSSB/ISSB climate-disclosure line items filed	z -score
<i>CarbonIntensity</i>	Scope 1+2 emissions / sales (industry-benchmarked)	ratio
<i>DisclosureCost</i> a	Assurance/audit-fee ratio; reporting-complexity proxy	ratio
<i>DistressCost</i> ϕ	Composite of leverage, Altman-Z, credit rating, and <i>CFvol</i> (proxy for the latent distress-cost motive; used in H5)	index
<i>Size</i>	$\ln(\text{total assets})$	$\ln$ KRW
<i>Leverage</i>	Total debt / total assets	ratio, winsor.
<i>MarketToBook</i>	Market value of equity / book value	ratio, winsor.
<i>ForeignSales</i>	Foreign sales / total sales	ratio
<i>CFvol</i>	12-quarter rolling s.d. of operating cash flow / assets	ratio, winsor.

Table 6: Variable definitions. Continuous variables are winsorized at the 1st/99th percentiles; ratios with denominators near zero (e.g. *FXHedgeRatio* for firms with negligible foreign sales) are set missing and not truncated, and the affected firms enter the FX regressions only. *CommodityHedgeRatio* is defined only on the commodity-exposed subsample (Section 6.3) to avoid the COGS-as-exposure contamination that afflicts a panel-wide construction. *DisclosureCost* is residualized on size, an internal-control-quality index, and industry before use (Section 4), since the raw proxy is entangled with risk-management capability.

Outcome	N	Not-yet	Adjacent cohort	Threshold-local	Placebo
Hedge-acct. designation	2,609	−0.005	−0.038	−0.019	+0.060
Derivative use	3,350	[−0.068, +0.060] −0.050	[−0.108, +0.033] −0.079	[−0.114, +0.068] −0.055	[−0.052, +0.175] +0.047
ln(1+FX-fwd mentions)	3,350	[−0.113, +0.013] +0.092	[−0.156, −0.015] +0.040	[−0.129, +0.019] +0.158	[−0.048, +0.135] +0.102
		[−0.037, +0.230]	[−0.122, +0.170]	[+0.003, +0.323]	[−0.107, +0.322]

Table 7: Governance-mandate estimates, OpenDART panel (380 firms, 3,420 firm-years, fiscal 2016—2024). Group-time ATTs under three comparison-set rules; 95% CIs from firm-cluster pairs bootstrap (999 replications). Placebo: fake $g=2019$ assigned to the 2024 cohort over 2016—2023 (true effect zero by construction).

Derivative	Predicted	Measured	Verdict
$\partial h_f^*/\partial \lambda$	+	+0.244	pass
$\partial d^*/\partial \lambda$	$\text{sign}(\phi - \lambda e^{-kd^*})$	−0.065 ($\phi - \lambda e^{-kd^*} = -0.412$)	pass (boundary)
$\partial h_f^*/\partial a$ (H(sub))	+	+0.190	pass
$\partial d^*/\partial a$	−	−1.346	pass
$\partial h_f^*/\partial p_f$	−	< 0	pass
$\partial h_f^*/\partial \sigma_f$	+	> 0	pass
$\partial h_f^*/\partial \rho$	sign of (24)	matches	pass

Table 8: Comparative-statics verification. Signs obtained by re-solving the full equilibrium (closed form, including the d^* feedback) under $\pm$ shocks, checked against the analytic predictions of Section 3.7. The disclosure/stringency derivative is checked against the derived boundary (22), not against a fixed sign.

Quantity	Value	Note
True ATT (model)	0.0621	average realized effect on h_f
Callaway—Sant’Anna ATT	0.0622	error $+5 \times 10^{-5}$
TWFE ATT	0.0549	error -7.2×10^{-3} (biased)
$\max \theta_\ell $, $\ell < -1$	0.00029	pre-trend (parallel-trends check)
H2: adoption DiD coefficient	0.706	positive (hedging-level channel)
H3: ATT carbon vs non-carbon	0.0717 vs 0.0518	stronger where climate risk larger

Table 9: Estimator recovery on model-generated staggered panels. The heterogeneity-robust group-time ATT recovers the true effect; two-way fixed effects is biased downward under dynamic effects and staggered timing, motivating the estimator choice of Section 5.3.

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